



Catchment Overview: River Llynfi

A report prepared on behalf of Wye Salmon Association (WSA)

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Executive Summary

The objective of this report is to review the current status of the Llynfi catchment¹, primarily with respect to phosphorus pollution. A key aspect of this is to leverage the citizen science data now becoming available. This data is generally collected with a significantly higher sampling frequency than the compliance data collected by NRW. Despite not having the measurement traceability required for compliance purposes it is, nevertheless, an information-rich dataset, better suited to monitoring trends and identifying sources of pollution.

The Llynfi catchment has a long history of eutrophication. In particular, Llangorse Lake in the upper reaches of the Llynfi has been much studied – largely out of concern for the tourist industry centred on the lake. However, the various recommended mitigation options have either not been implemented or have been ineffective.

Over the last few decades, significant efforts have been made by the water industry to divert sewage effluent from the upper catchment and to apply phosphorus stripping technology in the lower catchment. Sadly, there appears to have been little progress on agricultural pollution. Recent NRW compliance data (*Hatton-Ellis & Jones, 2021*) shows the Llynfi system to be approximately 3x the relevant SAC EQS targets for reactive phosphorus. NRW data published by *Goldsmith et al. (2018)* suggest that reactive phosphorus levels on Llangorse Lake are similar; moreover, there has been no significant change at the lake since the earlier report by *May et al. (2008)*.

The methodology of this report makes use of two complementary techniques: source apportionment modelling, as used in SAGIS, and load apportionment modelling which is a mathematical method of interrogating river water quality measurement data. Together, the two techniques tell a consistent story which is in good agreement with available data.

The report presents a simple source apportionment model for the Llynfi catchment, not dissimilar to earlier work by *May et al. (2008)* on Llangorse Lake. The model shows that the upper sub catchment is dominated by agriculture pollution, with 89% of river phosphorus load coming from diffuse agricultural sources. The remaining 11% originates from rural septic tanks. Using public-domain data for catchment land use and stock densities, the model suggests that 33% of the river phosphorus load comes from pastureland, with a further 34% from the sheep grazing on it; cattle contribute 11% of the total. The source apportionment model shows that the reactive phosphorus levels observed in the rivers of the Llynfi catchment follow in a very straightforward way from current agricultural practices.

In the lower sub catchment, below the Ennig confluence, there are three sewage treatment works; the largest, Talgarth has had phosphorus-stripping in place since 2015. Combined, these sources contribute 7% of the total phosphorus load in the lower Llynfi, with PTPs making up a further 2%. Whilst such annual average numbers may seem low, these ‘fixed flow’ sources become dominant at low summer flows when there is much reduced dilution of their effluent. The planned AMP8 upgrades to two of these plants are welcomed and our model suggests that these should reduce the total STW phosphorus load in the lower Llynfi by almost a factor of 3.

CSO spill data for 2020 has been analysed. This analysis shows that 4 CSOs need investigation by DC-WW. Whilst such spills are minor when viewed on an annual average basis, they can nevertheless be very significant pollution events at the time of the spill. This is especially true when the receiving water

¹ *Llynfi catchment* refers to the entire catchment of the river above its Wye confluence. Other geographical terms used in this report are defined [here](#).

happens to be in a low flow condition. Since we have no data concerning the timing of CSO spills, this remains an issue of high concern for these small rivers.

The load apportionment modelling presented here shows very clearly how reactive phosphorus levels rise as river flowrate – and hence dilution – reduces. A WSA citizen science RP dataset from Pipton Bridge on the lower Llynfi has been analysed in this way. The impact of Talgarth STW and Felindre STW are visible. Moreover, the fixed-flow coefficient extracted from this data is an excellent match for the source apportionment model presented for these STWs.

A second RP dataset from Llynfi Bridge at Llangorse has been analysed in similar fashion. Somewhat unexpectedly, it also shows a strong fixed-flow signal, equivalent in magnitude to treated effluent from several hundred people. There are no DC-WW STWs in the area, nor are there any licensed PTPs. The source of this signal is un-known and requires investigation.

We review the current River Wye SAC Nutrient Management Plan. Proposed measures to address the excessive phosphorus levels found within the Llynfi catchment appear woefully inadequate. Indeed, many of them are the same ones applied over the last thirty years, which have self-evidently failed to deliver any improvement. The rivers of the Llynfi catchment exceed SAC EQS targets for reactive phosphorus by a factor of 3-4x. This is not a marginal failure.

If compliance is to be achieved with SAC EQS targets for reactive phosphorus, then a paradigm change in agricultural land usage will be required; however, there is a lack of recent data for soil phosphorus concentrations within the catchment. Based on 2007 data, we estimate that legacy phosphorus would take 10 years to decay to best practice recommended levels, using typical crop off-take rates. The lack of data on soil phosphorus concentration is a matter for concern. It is axiomatic that this is a key driver of phosphorus export to the river and, as such, soil phosphorus concentration must be monitored.

This report also refers to agronomic modelling by the University of Lancaster which suggests that the entire Wye Catchment is applying 34% more phosphorus to land than is required for crop offtake. Thus, it is likely that soil phosphorus levels are currently increasing and have thus increased substantially since 2007. Because most of the excess phosphorus is sourced from livestock manure, this situation is likely to pertain within the livestock dominated Llynfi catchment.

This report makes several recommendations for action by key actors. Of particular importance, it is suggested that NRW feed back to the Welsh Government that significant changes to agricultural practices within the catchment are required if SAC EQS compliance is to be achieved. A few advisory farm visits and buffers strips are not going to have anything like the required impact. NRW should develop a credible improvement plan which takes full account of the magnitude of the task. This needs to include (a) a strategy for drawing down legacy phosphorus to an agreed environmentally sustainable level, and (b) ensure that future phosphorus inputs to land within the Llynfi catchment remain in balance with crop needs. It thus inevitably requires the creation of a system to monitor soil phosphorus levels.

Introduction

In line with its 'bottom up' approach to the use of its Citizens Science (CS) data, Wye Salmon Association (WSA) identified the River Llynfi and its tributaries (Ennig, Dulas & Felidre) as a potential exemplar, shining a light on identifying specific action tailored to one failing catchment, not just with significance to the widely publicised and failing River Wye SAC, but also demonstrating the potential merits of CS engagement more widely in Wales.

During the last twelve months, WSA volunteers along with Gwent Angling Society (GAS) members have participated in monitoring phosphate concentration in the Rivers Wye and Llynfi. This report makes use of data collected over the period July 2020 – August 2021. Two reports on the monitoring (*Gwent Angling Society, 2021a & 2021b*) confirmed NRW's findings from 2017-19 (*Hatton-Ellis & Jones, 2021*) that the Llynfi catchment fails to meet its SAC EQS target for phosphorus. What action NRW may have taken since 2019, or plans to take, to mitigate SAC non-compliance in the Llynfi catchment, is not known; nor is any such action recorded in the *River Wye SAC Nutrient Management Plan: Phosphate Action Plan 2021*.

WSA data identified a number of conflicting and unexplained results emanating at various points in the course of the river's flow from Llangorse Lake.

The objective of this report is to examine the continuous, background pollution which impact the Llynfi catchment. In the rural setting of this catchment, surface water pollution originates from two principal sources: agriculture and human sewage. We focus primarily on phosphorus pollution as a key indicator, although it is understood that other concomitant pollutants can be equally, or even more harmful to the river.

Two complementary approaches have been utilised. One is a source apportionment model, in the manner of SAGIS, where we estimate the input phosphorus loads within the catchment. The second is load apportionment modelling, where we analyse citizen science phosphorus concentration data to see what the river is telling us. The two approaches tell a consistent story of a catchment which is heavily agriculture-dominated on an annual mean basis, but where the lower river becomes sewage dominated during low flow conditions.

A secondary objective has been to build a simple, open-source model for the catchment, as opposed to the esoteric world of SAGIS. Clearly, we do not have access to much of the detailed data which is available to the water companies and environment agencies. So, we must inevitably make some assumptions and approximations along the way. Consequently, this document is version controlled, in order that it can be updated as new information emerges.

Catchment overview

Principal data sources for this section are:

- [National River Flow Archive \(ceh.ac.uk\)](http://ceh.ac.uk) (NRFA)
- [UK Soil Observatory \(bgs.ac.uk\)](http://bgs.ac.uk) (UKSO)
- [Farming Facts and Figures, Wales 2021 \(gov.wales\)](http://gov.wales) (FFFW)

Both NRFA and UKSO are managed by the UK Centre for Ecology and Hydrology.

The river

The river Llynfi rises on the norther slopes of the Black Mountains and flows via Llangorse Lake, in a generally northerly direction, to join the river Wye at Glasbury; a total distance of around 20km. The fall from source to confluence is approximately 95m. In the lower reaches, just outside the town of Talgarth, the Llynfi is joined by the Dulas Brook which, along with its tributary the Tryffrwd Brook, drains the western side of the catchment. Just below this point, the Ennig joins from the east after flowing through Talgarth. Further downstream near Three Cocks, the Felindre and Tregoyd brooks also join from the east. A map of the catchment is shown in Figure 1 below.

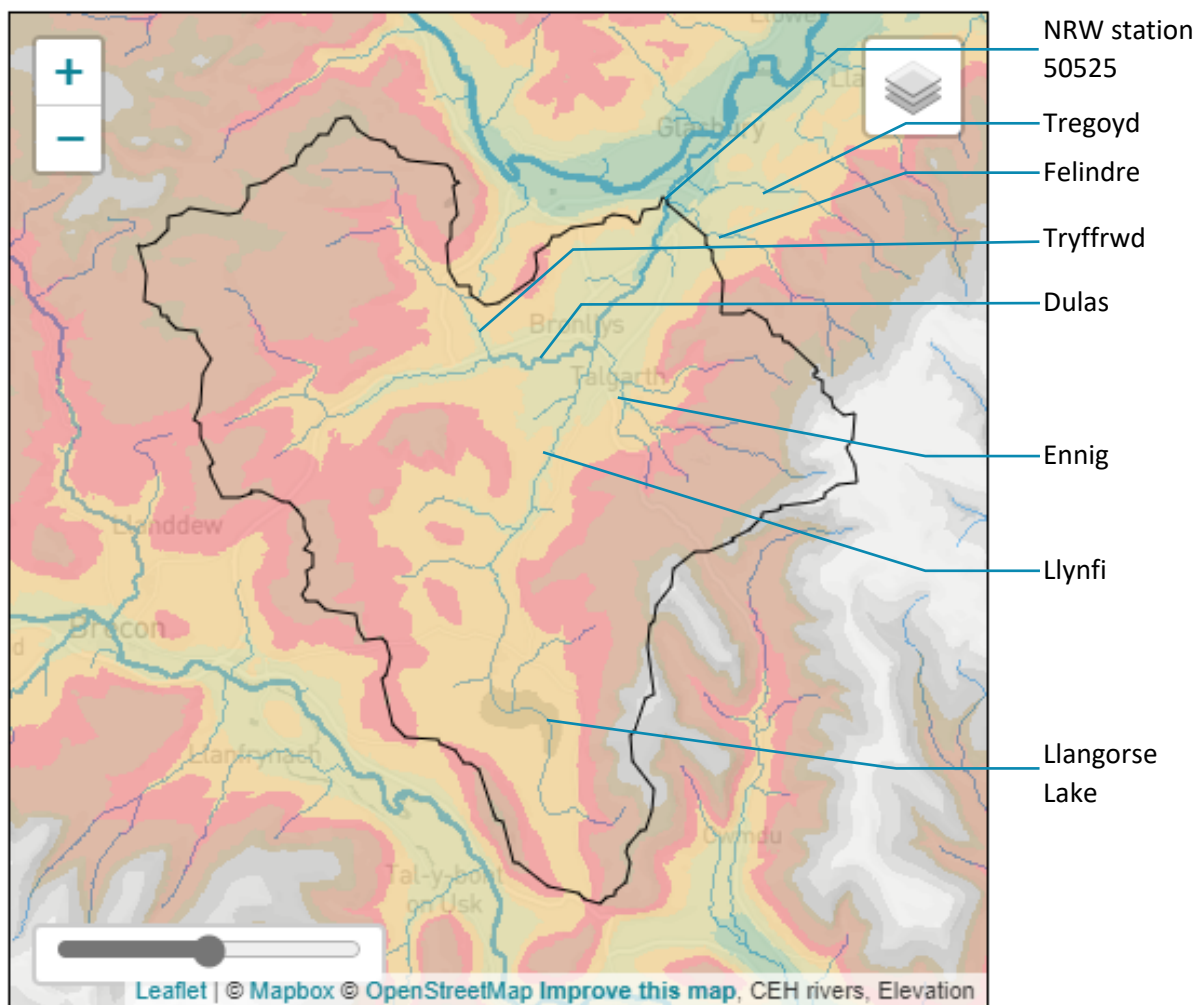


Figure 1: topographical view of the Llynfi catchment (from NRFA)

The majority of the catchment is defined with respect to the NRW flow gauging station, ref. 55025, at the village of Three Cocks (SO166375), approximately 2km above the Wye confluence (note however that the Felindre and Tregoyd brooks join below this point). The total catchment area above this measurement station is 132 km². Average annual rainfall is 1000mm, ranging from 1150mm in the hills around the edges of the catchment, to 850mm near Three Cocks (NRFA).

Key flow parameters for the river at the Three Cocks measurement station are as follows:

Base flow index ²	0.57
95% exceedance flow (Q95) ³	0.16 m ³ /s
50% exceedance flow (Q50) ⁴	1.33 m ³ /s
Mean flow	2.30 m ³ /s

Table 1: Llynfi flow parameters at Three Cocks, taken from NRFA

Based on the mean flow, the Llynfi carries 55% of the rain landing on its catchment; this is often referred to as the hydrologically effective rainfall (HER), the fraction of which is calculated in Box 1 below. The remaining 45% is lost to evaporation and transpiration. Such numbers are not untypical of the wider Wye catchment.

Box 1: effective rainfall calculation

Volume of rain	= average rainfall x catchment area = 132 x 10 ⁶ m ³ /annum = 4.19 m ³ /s
HER fraction	= Llynfi mean flow / volume of rain = 2.30 / 4.19 = 55%

It will be useful later to know roughly how much water flows in the various tributaries at different points within the catchment. In the absence of flow measuring stations, we must estimate these. To this end, we have first estimated the areas of the sub-catchments of interest using the topographical mapping and then calculated the HER, in the same manner as Box 1. Results are as follow:

Sub catchment	Area	HER	Mean Flow	Q95 Flow
Dulas, source to Llanfilo	25 km ²	0.44 m ³ /s	0.44 m ³ /s	0.03 m ³ /s
Dulas, source to Llynfi confluence	40 km ²	0.70 m ³ /s	0.70 m ³ /s	0.05 m ³ /s
Ennig, source to Llynfi confluence	22 km ²	0.38 m ³ /s	0.38 m ³ /s	0.03 m ³ /s
Llynfi, above Llynfi Bridge Llangorse	23 km ²	0.40 m ³ /s	0.40 m ³ /s	0.03 m ³ /s
Llynfi, Llangorse to Dulas conf.	36 km ²	0.63 m ³ /s	1.03 m ³ /s	0.04 m ³ /s
Llynfi, Dulas conf. to Three Cocks	11 km ²	0.19 m ³ /s	2.30 m ³ /s	0.16 m ³ /s
Felindre, source to Llynfi confluence	10 km ²	0.17 m ³ /s	0.17 m ³ /s	0.01 m ³ /s
Tregoyd, source to Llynfi confluence	7 km ²	0.13 m ³ /s	0.13 m ³ /s	0.01 m ³ /s

Table 2: estimated flow values for some key sub-catchments, HER calculated as 55% of 1000mm p.a. average rainfall; Mean Flow values are cumulative

² Base flow index is a measure of 'flashiness' of a river. A chalk-stream would have a value close to 1. The Wye catchment varies from 0.34 on the Elan, to 0.65 on the Lugg.

³ Q95 is the flow which is exceeded for 95% of the time. It represents the river 'on its bones' and is important for worst-case dilution considerations.

⁴ Q50 flow is exceeded 50% of the time. It is arguably closer to what we would intuitively call 'normal flow' than is the mean flow, which is influenced by brief but high spate events.

Q95 values above are calculated as a fixed ratio to mean flow. This is very approximate, not least because of the damping effect of Llangorse lake noted below. But it gives us a first order estimate, so it's better than nothing.

At this point, we should define some terminology:

- The *Upper Llynfi* refers to the river above Llangorse Lake.
- The *Middle Llynfi* refers to the river from Llangorse down to the Ennig confluence.
- The *Lower Llynfi* refers to the river from the Ennig confluence to the Wye confluence.
- The *Upper Sub Catchment* refers to everything above Talgarth, comprising the Dulas, Ennig and Llynfi down to the Ennig confluence; it thus includes 121 km² of the 132 km² catchment area above the Three Cocks gauging station.
- The *Lower Sub Catchment* refers to catchment area below the Ennig confluence; it includes the 11 km² balance of the Three Cocks catchment (land drained by the *Lower Llynfi* between the Ennig confluence and Three Cocks), plus the Felindre and Tregoyd sub catchments which add a further 17 km².
- The full *Llynfi Catchment* is thus the sum of the *Upper* and *Lower* sub catchments: approximately 149 km² in total.

Llangorse Lake

The presence of the lake in the headwaters of the Llynfi is a significant factor for nutrient concentrations in the river Llynfi. Llangorse has a long history of eutrophication and has been much studied (*May et al., 2008; Goldsmith et al., 2018*).

The lake is fed by seven streams, with a combined catchment of 23 km², representing 17% of the Llynfi total. Mean outflow into the Llynfi was estimated by *May et al.* as 0.76 m³/s, representing 30% of the Llynfi total. The fact that this is disproportionate to the catchment area was discussed by *May et al.* who found that the 'excess' flow is measurable on the input streams and is particularly notable on the Upper Llynfi. The authors suggested possible explanations, but the issue was not investigated further.

While noting this effect, we have not included it in the calculations to follow because we do not know that the upper catchments of the other tributaries do not have similar complexities. We are aware of this at Llangorse purely because the area has been heavily studied.

Several of the input streams dry up in dry weather and *May et al.* reported that most of the input flow over the period of their study came from Nant Cwy (18%) and the Upper Llynfi (67%).

Retention time for water in the lake was estimated as 55 days. The presence of the lake will undoubtedly serve to damp down the flashiness of the upper Llynfi.

In the 1970s, annual mean total phosphorus (TP) in the lake was 0.16mgP/l. Following diversion of the Llangorse and the Bwlch STWs away from the lake, this reduced to 0.12 mgP/l by the mid '90s - still massively above the SAC TP target for Llangorse in 2008 of 0.035 mgP/l. The work of *May et al.* recommended that a further 15% reduction of TP was required to mitigate against algal blooms. However, they established that for the most part the algal blooms were limited by nitrogen rather than phosphorus. More recent work by NRW (*Goldsmith et al., 2018*) showed annual mean TP levels in the lake over the period 2014-18 were broadly similar to the 2008 result. It would appear that the various mitigation options discussed by *May et al.* either were not implemented or were ineffective.

Geology

The headwaters of the Llynfi drain the old red sandstone of the Black Mountains. Lower reaches flow over old red sandstone marls. Thus, the whole catchment sits on low permeability bedrock, meaning that all drainage water, and pollutants therein, flow to the river.

Soil within the catchment is typical of much of Powys, less than a metre deep on the uplands and increasing into the valley bottoms. In only a few scattered areas, predominantly below Talgarth, do we find deeper soil.

According to UKSO, soil bulk density averages at 1.03 kg/l across the catchment. UKSO quote soil P measurements (Olsen-P) as mg/kg, whereas much of the agricultural world seems to prefer mg/l. We can readily convert the mg/kg value to mg/l by multiplying by bulk density in kg/l.

Box 2: Olsen-P

Olsen-P is a standard measure of soil phosphorus, developed by Olsen, and widely used in agriculture. It is the main method used in England, Wales and Northern Ireland and the basis of the soil index for P, as used by farmers.

It does not measure total phosphorus but measures the fraction which is extracted by the chemistry used. It is based on the molybdenum blue method, using sodium bicarbonate extraction at pH 8.5. It is generally regarded as approximating to plant-available phosphorus.

Land use

The Llynfi catchment is typical of much of Powys, being dominated by grassland. Figure 2 below shows the land use across the catchment. The numerical breakdown is as follows:

Woodland	12%	1,554 ha
Arable/horticulture	9%	1,247 ha
Grassland	72%	9,540 ha
Mountain, heath & bog	3%	360 ha
Urban	2%	236 ha

Table 3: land use in the Llynfi catchment

Thus, farming in the catchment is dominated by livestock: sheep on the uplands and some cattle in the lower areas. The small amount of arable farming is mostly concentrated in three areas, the largest of these in the lower valley around Talgarth and Three Cocks.

Livestock numbers for Wales are available in *Farming Facts & Figures, Wales 2021*. In the absence of specific data, we can estimate the head of livestock in the Llynfi catchment from this by assuming the same stock densities for grassland, as follows:

	Wales	Llynfi catchment
Area of grassland	1,006,000 ha	9,540 ha
Total sheep	8,989,000	85,000
Total beef cattle	162,000	1,500
Total dairy	252,000	2,400

Table 4: livestock estimates for the Llynfi catchment

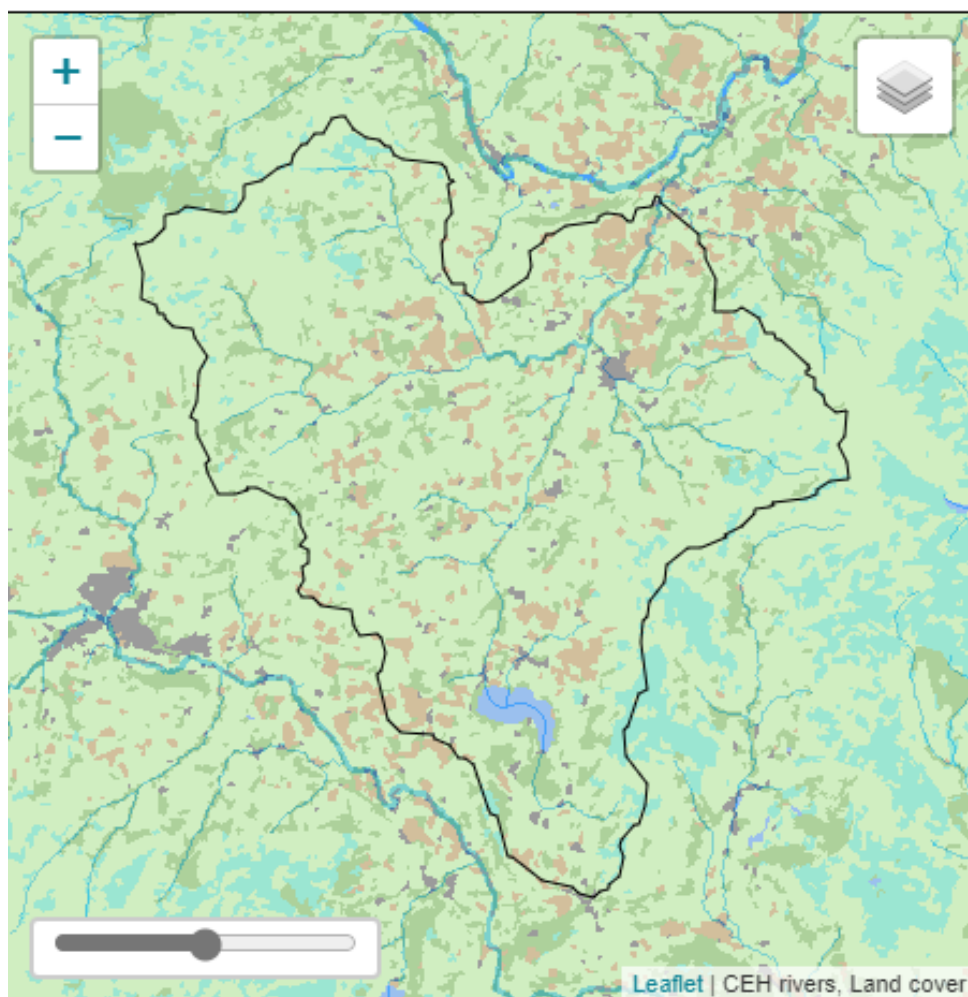


Figure 2: land use map of the Llynfi catchment; grassland is light green; woodland is dark green and arable is tan

Legacy phosphorus

Sadly, UKSO carries no current data for soil Olsen-P. However, they do provide results for 1998 and 2007, which we have sampled at 16 points across the catchment. The mean values are as follow:

1998 mean Olsen-P	40 mgP/l
2007 mean Olsen-P	29 mgP/l

Table 5: mean soil Olsen-P values

It is known that most of England and Wales was subjected to major over application of phosphorus in the post war years, 1950s – 1980s (*Phosphorus and freshwater eutrophication narrative, 2019*). It is perhaps encouraging that the results above show a decline between 1998 and 2007. Whether this decline has continued is a moot point. In more recent times, plateauing yields and economic factors have mitigated against profligate use of artificial fertilisers. However, it is known that many farmers have typically applied organic manures according to crop nitrogen requirements, which inevitably overdoses phosphorus. For lack of any other data, therefore, we shall use the 2007 figure in our calculations.

According to The Soil Association (*Soil management on organic farms, 2003*) and the Agriculture & Horticulture Development Board (*RB209: Nutrient management guide*), target phosphorus levels (for most soil types, and most crops in the case of arable) should be as follow:

Target Olsen-P pasture	18 mgP/l
Target Olsen-P arable	25 mgP/l

Table 6: best practice targets for Olsen-P

As noted in Box 2 above, Olsen-P is not a measure of the total phosphorus in the soil. So, at this point, we need to determine how Olsen-P relates to total phosphorus. *RB209* informs us that ‘it can take 200-300 kgP₂O₅/ha⁵ to raise Olsen-P by 10 mg/l’. In order to compare these two quantities, we need a number for soil depth. Now the accepted view (*Heckrath et al., 1995; Smith et al. 1998; Rashmi et al., 2020*) is that phosphorus accumulates only in the upper 200-300mm or so of the soil – irrespective of total physical soil depth; there is minimal accumulation in sub-soil. With this information, we can relate Olsen-P to soil total phosphorus as follows:

Box 3: Olsen-P as a fraction of total phosphorus

Effective soil depth:	250 mm
Effective soil volume:	10,000 m ² /ha x 0.250 m = 2,500 m ³ /ha
Applied phosphorus:	250 kgP ₂ O ₅ /ha = 110 kgP/ha
Applied P concentration:	(110 x 10 ⁶ mgP) / (2,500 x 1,000 l) = 43.7 mgP/l
Olsen-P increment:	10 mgP/l
Olsen-P / Total P:	23%
(Note that this can only ever be a first-order estimate; in practice the ratio will vary from field to field, depending on a multiplicity of factors. I will round this to 25% for the following calculations.)	

With this information, we can now estimate how much excess phosphorus is in the Llynfi catchment. We do this separately for the two categories of land use as follows:

	Excess Olsen-P	Excess Total P	Excess as P ₂ O ₅	Area	Total P ₂ O ₅
Arable	4 mg/l	48 kgP/ha	110 kg/ha	1,247 ha	137 tonnes
Grassland	11 mg/l	88 kgP/ha	202 kg/ha	9,540 ha	1,922 tonnes

Table 7: excess phosphorus within the Llynfi catchment. Excess Total P is calculated using Olsen-P as 25% of Total P and soil depths of 200mm for grassland, 300mm for arable land – see Box 4 for details.

In terms of nutrient load on the river, the problem to solve in this catchment is clearly the grassland, which accounts for 93% of the legacy phosphorus (Total P₂O₅ in Table 6). The next question is how to run down this legacy phosphorus. *RB209* teaches us that a typical annual crop off-take for pasture is 20 kgP₂O₅/ha. This suggests that it will take at least 10 years to run down the 202 kgP₂O₅/ha excess, assuming no further phosphorus addition. Based on export coefficients presented later in this report,

⁵ P₂O₅ is the chemical form of phosphorus often found in synthetic fertilisers. Therefore, in agriculture, kg P₂O₅ has become the preferred ‘unit of currency’ for phosphorus (P makes up 44% of P₂O₅ by mass.)

without crop off-take (that is, relying entirely on leaching to the river) legacy phosphorus run-down would take around two hundred years.

A preliminary analysis by the University of Lancaster, presented to the Wye Nutrient Management Board on 19-January 2021 by Dr Shane Rothwell, suggests that legacy phosphorus is still increasing across the entire Wye catchment at a rate of 2000 tonnesP/year. That is 8 kgP/ha/year, excluding rough grazing. Every year the land is receiving 34% more phosphorus than is required to balance agricultural offtake. Of the surplus, 77% is coming from livestock manure, with sheep and cattle at 26% each. Thus, these numbers are undoubtedly indicative of the situation in the Llynfi catchment.

In examining legacy phosphorus, we are frustrated by two issues:

- Firstly, no comprehensive soil P surveys appear to have been carried out since 2007. Certainly, none are reported on CEH websites (the absence of up-to-date data was also commented on by Dr Rothwell).
- Secondly, the data that we do have report only Olsen-P, meaning that we must rely on an assumed ratio of Olsen-P to total P. But it is well known that this ratio is heavily dependent on local soil type, so the procedure adopted here is less than ideal.

The current situation is one where farmers are encouraged to operate nutrient management plans for land, underpinned by periodic soil nutrient measurements; this is encouraged by COGAP guidance and by RB209. At least some of the farmers within the Llynfi catchment must be doing this. Surely, this data can be systematically collected by the responsible agency.

The current absence of soil phosphorus data is bewildering in the context of an SAC which is failing water quality targets. It is axiomatic that soil phosphorus concentration is a key driving factor for river phosphorus enrichment and thus requires regular monitoring. How can NRW hope to manage nutrient concentrations in rivers without measuring nutrient concentrations in the soils draining into the rivers?

Box 4: Calculation of Excess Total P in Table 6

Land use	Grassland	Arable
Effective soil depth:	200 mm	300 mm
Effective soil volume:	2,000 m ³ /ha	3,000 m ³ /ha
Excess Olsen-P:	11 mg/l	4 mg/l
Excess Total P:	11 / 25% = 44 mgP/l = 44x10 ⁻³ kgP/m ³ = 44x10 ⁻³ x 2,000 kgP/ha = 88 kgP/ha	4 / 25% = 16 mgP/l = 16x10 ⁻³ kgP/m ³ = 16x10 ⁻³ x 3,000 kgP/ha = 48 kgP/ha

Sewage effluent

Principal data sources for this section are:

- [Raw sewage in our rivers | The Rivers Trust](#)
- Dwr Cymru – Welsh Water: list of assets provided to WSA via FoI request
- [Wales / Cymru \(United Kingdom\): Districts & Settlements - Population Statistics, Charts and Map \(citypopulation.de\)](#)

Human sewage contributes a variety of pollutants to our rivers. Residual organics (BOD), nutrients (phosphorus, etc.), pharmaceuticals, pesticides, cleaning chemicals, harmful bacteria; the list goes on. The levels at which these substances get into rivers is largely an economic choice, a cost-benefit decision. There are no zeros here. It's about agreeing an acceptable level.

A key aspect of these sources is that they are broadly constant, irrespective of rainfall. Therefore, their effects are amplified by reduced dilution through the summer months. Even CSOs, which in theory are only supposed to discharge in heavy rain, in practice often discharge in dry conditions due to blockages and other malfunctions, or even by design as per the recent Southern Water legal case.

The following sections are structured around the different categories of source.

Sewage treatment works

Introduction

STW loads vary. Some serve only residential properties; others might include commercial and industrial effluent. In consequence, the water industry has adopted the concept of Population Equivalent (PE) as a standardised unit of load. An input load of 1 PE is commonly defined (within the UK and EU) as follows:

Liquid	180 l/day
Solids	250 g/day
BOD	60 gO ₂ /day
N	12 gN/day
P	2.5 gP/day

Table 8: input loads per PE

Within the STW, some of the nutrients are taken out with the solids and some remain in solution. In this report I will focus primarily on phosphorus and treated sewage effluent. A typical output stream per PE for an STW with traditional primary and secondary treatments (no phosphorus stripping) would be:

Liquid	180 l/day	
P	1.44 gP/day ⁶	= 8.0 mgP/l

Table 9: outputs per PE; note that P is total phosphorus

When phosphorus stripping (a tertiary treatment) is added, the P concentration will reduce to below 2 mgP/l. Indeed, modern systems are capable of achieving well below 1mgP/l where needed. The phosphorus stripping process incurs significant running costs, so the phosphorus target for an individual STW will be an economic balance between such costs and the environmental impact.

⁶ This quantity is often referred to as an 'export coefficient'; it can otherwise be stated as 0.54 kgP/PE/year. We will meet several more export coefficients later in this report.

Current emissions

There are four sewage treatment works (STW) within the Llynfi catchment as follows:

STW Name	Serves	Discharges to	Input PE
Aberllynfi STW	Three Cocks	Llynfi (lower) via Tregoyd Bk.	295
Llanfilo STW	Llanfilo	Dulas (upper)	54
Talgarth STW	Talgarth & Bronllys	Llynfi (lower)	2,422
Felindre STW	Felindre	Llynfi (lower) via Felindre Bk.	160

Table 10: STWs within the Llynfi catchment; Input PE values are DC-WW data

As can be seen from the PE values, Talgarth (pop. 1295) and Bronllys (pop. 577) together represent by far the largest settlement within the catchment. Talgarth STW is currently the only one to have had phosphorus stripping fitted; this was added in the AMP5 investment cycle (2010-15), with an annual average effluent limit set at 2.0 mgP/l. The other three STWs have just standard primary and secondary treatments. Further investments in phosphorus stripping are planned within the AMP8 investment cycle (2025-30); we shall discuss these later.

The situation at Llangorse Village (est. pop. 360) deserves discussion. A discharge from Llangorse STW originally flowed into the lake. However, this was diverted to the Llynfi outflow in 1981, in order to reduce the nutrient load on the lake. This appears to have been the situation at the time of the work reported by *May et al. (2008)*. Currently, Llangorse Village sewage is piped down the valley to Talgarth for processing; as such, it is included within the stated input PE at Talgarth. It flows via a pumping station located adjacent to Llynfi Bridge, immediately to the west of Llangorse Village, complete with a storm overflow (the overflow is not shown on Rivers Trust mapping, so presumably is un-permitted).

The other notable feature at Llangorse is the large seasonal tourist population. Aerial mapping shows around 230 chalets and fixed caravans located adjacent to the Llynfi at Llangorse, plus a further 85 on the other side of the village; with an average occupancy of 2.5, this equates to 788 people. To this, we should add an indeterminate number of campers and B&B guests. So, it is easy to imagine a peak tourist population of 1000 in the immediate vicinity of Llangorse Village. Stated as an annual average load, this would equate to maybe 250 PE. It is entirely unclear how many of these are on mains drainage and how many are relying on on-site treatment.

We can now calculate the effluent from the four STWs discharging into the rivers, using the PE definition in Table 9, or (in the case of Talgarth) the applicable annual average P and multiplying by the PE load for each STW. The result is the data shown below:

STW Name	Effluent P conc.	Liquid flow	P Load
Aberllynfi STW	8.0 mgP/l	0.61 l/s	0.42 kgP/day
Llanfilo STW	8.0 mgP/l	0.11 l/s	0.08 kgP/day
Talgarth STW	2.0 mgP/l	5.05 l/s	0.87 kgP/day
Felindre STW	8.0 mgP/l	0.33 l/s	0.23 kgP/day
		6.11 l/s	1.60 kgP/day

Table 11: STW effluent streams within the Llynfi catchment; P load is total phosphorus

Let's look at the phosphorus load graphically:

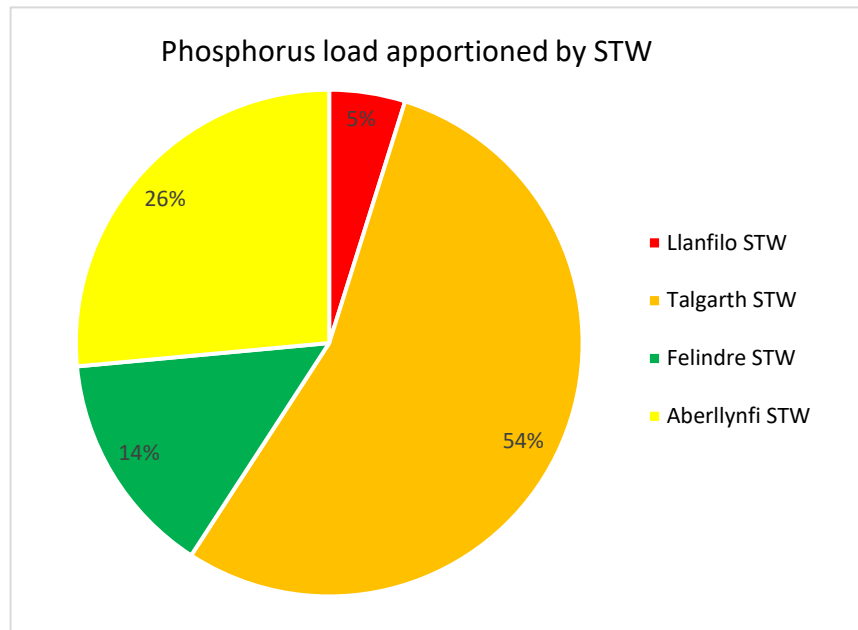


Figure 3: catchment phosphorus load apportioned by STW

This shows that Talgarth, despite its P-stripping treatment, is still contributing over a half of the total phosphorus load, followed by Aberllynfi with a quarter. Llanfilo and Felindre are less significant at the catchment level.

In the lower river, below the Tregoyd confluence, all four flows are additive. So, we have 1.6 kgP/day flowing through this section of river. In the table below, we also look at the impact on three key sub catchments. We will take the following totals through to our later summation:

Sub catchment	STWs	P load
Dulas Brook	Llanfilo	0.08 kgP/day
River Llynfi above Dulas confluence	-	0.00 kgP/day
River Ennig	-	0.00 kgP/day
Full Llynfi catchment	All	1.60 kgP/day

Table 12: daily average phosphorus load from STWs

Finally, we shall look at dilution factors during low summer flows, including on some smaller tributaries. Of particular interest is the reduced dilution in these conditions and the resulting river phosphorus concentrations. Key dilution metrics are shown in Table 13 below.

Given its high PE load, it is unsurprising to see Talgarth STW showing poor dilution at Q95 flowrate. However, the impact on river phosphorus concentration is mitigated somewhat by phosphorus stripping. That said, the predicted river phosphorus concentration of 0.06 mgP/l is still substantial, even as TP.

Perhaps of more concern in Table 13, we see Felindre and Aberllynfi STWs, despite being relatively small in an overall sense (Figure 3), also standing out with poor dilution and massive predicted phosphorus concentrations, resulting from the small summer flowrates in the Felindre and Tregoyd brooks. Essentially these brooks are being used as simple conduits to get the effluent down to the Llynfi; cheaper than installing a pipe.

Name	Effluent flow	Discharges to	Q95 flow	Dilution	P conc.
Llanfilo STW	0.11 l/s	Dulas, source to Llanfilo	30 l/s	0.4%	0.03 mgP/l
Talgarth STW	5.05 l/s	Llynfi, above Three Cocks	164 l/s	3.1%	0.06 mgP/l
Felindre STW	0.33 l/s	Felindre	12 l/s	2.8%	0.22 mgP/l
Aberllynfi STW	0.61 l/s	Tregoyd	9 l/s	6.9%	0.55 mgP/l

Table 13: dilution parameters for each STW; Effluent (liquid) flow is reproduced from Table 11; Q95 Flow for each section of river is taken from Table 2; Dilution is the ratio of Effluent flow to river Q95 flow; P conc. is the predicted contribution from each STW to total phosphorus concentration in the river (P Load from Table 11 divided by Q95 flow).

These Q95 numbers are particularly important, because they occur in the growing season; consequently, they have the potential to drive algal blooms and otherwise modify in-river plant populations.

The Q95 flow at Three Cocks is a real, measured value. But as noted earlier in connection with Table 2, the remaining Q95 flows are first order estimates only. Nevertheless, when we see predicted phosphorus concentrations a whole order of magnitude higher than the SAC EQS targets for RP (0.015 – 0.025 mgP/l), then this result cannot be waived away by uncertainties in the Q95 flow estimate.

At this point we should examine the proposals for future investment in the STWs within the Llynfi catchment.

Future investment plans

WSA have been advised of investment plans for the Wye catchment by Dwr Cymru – Welsh Water via a formal information request. The Llynfi catchment does not feature in the current AMP7 investment cycle, but there are some relevant developments in AMP8 (2025-30 delivery), as follow:

STW name	Cycle	Current P limit	New P limit	P reduction	P conc.
Talgarth WwTW	AMP8	2.0 mgP/l	0.6 mgP/l	0.61 kgP/day	0.02 mgP/l
Aberllynfi WwTW	AMP8	None	1.5 mgP/l	0.34 kgP/day	0.10 mgP/l
				0.95 kgP/day	

Table 14: DC-WW future investments in phosphorus stripping; P reduction is our calculation, based on PE load; P conc. is the resulting contribution to river P load at Q95 flow (cf. Table 12).

So, the AMP8 investment cycle will see a welcome reduction from Talgarth and Aberllynfi STWs, reducing the total STW P load from 1.60 kgP/day to 0.65 kgP/day. By 2030, the apportionment will be as shown in Figure 4 below. In terms of summer flows, the improvement is significant for the Llynfi below Talgarth STW, but less impressive for the Tregoyd brook, which will still have very high summer phosphorus levels. No improvement is planned for the Felindre STW.

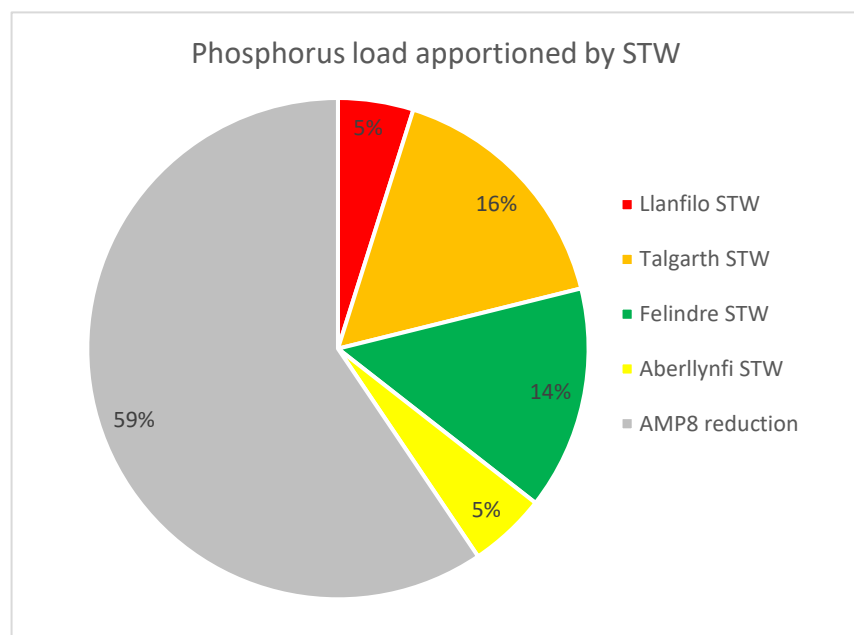


Figure 4: catchment phosphorus load apportioned by STW, post AMP8 investment

Septic tanks

Introduction

Septic tanks (STs) are a nightmare. They are numerous, often very old and largely undocumented, as noted by *Edwards* in his review paper on point source pollution. An excellent review of STs is to be found in *May et al. (2015)* and follow up work reported by *Broadhead (2016)*.

A septic tank is a simple two chamber system providing a *primary* treatment (analogous to the primary treatment in an STW). In the first chamber, solids settle to the bottom and scum floats to the top. The remaining liquid flows into the second chamber, from where it is routed to a soakaway in adjacent land – this being considered a type of *secondary* treatment. It is generally assumed that the effluent is ‘cleaned up’ as it percolates through the soil to the river, although there appears to be little evidence to back this up in the context of phosphorus. Indeed, the soil is known to have a finite capacity for phosphorus and one study showed that the phosphorus ‘plume’ extended 30m downstream of a soakaway and advanced by 1m per year. Thus, the performance of soakaways is known to degrade over time.

May et al. (2015) identified two routes for phosphorus to reach surface water. In dry weather, transfer from the soakaway occurs via seepage through the subsoil. In wet weather though, higher levels of phosphorus can be flushed from the soil and from the tanks. At such times, the soil around a soakaway can become a phosphorus emitter, rather than an absorber.

May et al. (2015) report studies of English catchments where only 3-10% of STs are consented. There is no reason to think that the situation in the Llynfi catchment is substantially different. Furthermore, other studies found the majority (80%) of STs appear to be poorly maintained, in the main not being emptied often enough. Old systems often discharged directly to surface water (against current regulations). Others had soakaways too close to surface water (again failing current regulations). Many Victorian systems even route roof runoff via the ST. These three issues all result in ‘hydraulic failure’

(overload) whereby untreated effluent bypasses the tanks and goes directly to discharge, increasing phosphorus output accordingly.

Current regulations stipulate that all new STs should be registered with NRW and there are appropriate design standards in place. Furthermore, direct discharge to surface water is forbidden without secondary treatment, which precludes simple STs.

Export coefficient

Given all the practical issues around STs and the lack of evidence to the contrary, the accepted view is to assume that all ST phosphorus eventually gets to the river. *May et al. (2008)* used an export coefficient of 0.4 kgP/PE/year in their Llangorse study. *May et al. (2015)* suggest that 0.54 kgP/PE/year is the consensus value in more current literature, a value which is not inconsistent with the findings of *Broadhead (2016)*. In other words, STs are found to have a similar phosphorus output per PE to STWs.

Current emissions

All houses and farms outside of the limited areas served by the four STWs must have septic tanks. *May et al. 2008* estimated that there were 1000 people in the Llangorse Lake catchment alone who were served by septic tanks. So, how many are in the full Llynfi catchment today?

A way to approach this problem is to look at regional population densities. Statistics based on census information are counted by ward. These don't neatly correspond to catchment boundaries, but the main ones which cover the Llynfi catchment are as follow:

Ward	Area (km ²)	Population	Pop. density
Llangorse	36	1100	36
Felinfach	55	645	12
Talgarth	55	1727	31
Bronllys	19	935	48

Table 15: population densities within the Llynfi catchment; data are 2020 estimates based on the 2011 census, as reported on the [City Population website](#)

Talgarth, Bronllys and Llangorse wards are all influenced by having relatively significant settlements within their boundaries which all have mains drainage. But it could be argued that Felinfach is typical of the rural catchment. It does not appear to include any large settlements or STWs.

On this basis, we can take the population density of Felinfach ward, multiply by the area of the catchment, and estimate the number of people on septic tanks. We can now multiply this by our export coefficient (0.54 kgP/PE/year = 1.44 gP/PE/day) to determine the catchment and sub catchment loads as follow:

Sub catchment	Area	Pop'n on STs	P load
Dulas Brook	40 km ²	469	0.68 kgP/day
River Llynfi above Dulas confluence	59 km ²	694	1.00 kgP/day
River Ennig	22 km ²	258	0.37 kgP/day
Full Llynfi catchment	132 km²	1548	2.23 kgP/day

Table 16: daily average phosphorus load from septic tanks; areas are from Table 2; 'P load' is calculated at 1.44gP/PE/day x 'Pop'n on STs'

Package treatment plants

Introduction

Once again, *May et al. (2015)* and *Broadhead (2016)* provide useful data.

Package treatment plants (PTPs) differ from septic tanks insofar as they incorporate a secondary treatment process and thus have tighter effluent specifications. The technology can be added onto the end of a standard ST or supplied as an integrated unit. This inclusion of secondary treatment allows PTPs potentially to discharge directly into surface water, subject to NRW licensing.

Unfortunately, the improved performance of PTPs relates primarily to BOD and suspended solids. Although some PTPs are capable of phosphorus reduction, this technology is not typically deployed. Standard PTPs appear to output only marginally less phosphorus than simple septic tanks. However, *Broadhead (2016)* suggests that direct discharging PTPs can often have higher phosphorus emissions to surface water than a septic tank when the effect of the ST's soakaway is included.

Export coefficient

In the light of the above, it seems reasonable once again to use an export coefficient of 1.44 gP/PE/day.

Current emissions

According to the Rivers Trust data, there are 18 licensed PTPs within the Llynfi catchment, all but 2 of these discharging directly to rivers. As discussed above, it is reasonable to assume that a discharge to a soakaway ultimately carries its load of nutrients to surface water; it just gets there more slowly.

Therefore, these are not excluded these from the analysis. The list is as follows:

Name	Discharges to	Discharge	PE	P Load
Gilfach	Llangorse Lake	Direct	3	0.004 kgP/day
Old Vicarage	Llangorse Lake	Direct	3	0.004 kgP/day
Pendre Courtyard	Llangorse Lake	Soakaway	10	0.014 kgP/day
Felinfach (Griffiths) STW	Dulas Brook	Direct	3	0.004 kgP/day
Felinfach (Griffiths) STW	Dulas Brook	Direct	3	0.004 kgP/day
Griffin Meadows Felinfach	Dulas Brook	Direct	3	0.004 kgP/day
Griffin Inn Felinfach	Dulas Brook	Direct	20	0.029 kgP/day
Pontybat Cottage	Dulas Brook	Direct	3	0.004 kgP/day
Llandefalle STW	Tryffrwd	Direct	3	0.004 kgP/day
Bronllys Hospital	Dulas Brook	Direct	100	0.144 kgP/day
Watson Cottage STW	Dulas Brook	Direct	3	0.004 kgP/day
PTP @ Bronllys Castle	Llynfi	Direct	20	0.029 kgP/day
STP serving Honey Café	Llynfi	Direct	20	0.029 kgP/day
Great Porthamel Farm ⁷	Llynfi	Direct	25	0.036 kgP/day
Septic tank @ Multiple	Felindre	Soakaway	20	0.029 kgP/day
Maes y Lade Tregoyd	Felindre	Direct	3	0.004 kgP/day
Old Railway Line G Centre	Felindre	Direct	20	0.029 kgP/day
Three Cocks Tregoyd	Tregoyd	Direct	20	0.029 kgP/day
STP Serving The Kennels	Tregoyd	Direct	10	0.014 kgP/day
			292	0.420 kgP/day

Table 17: licensed PTPs within the Llynfi catchment; PE are our estimates; P Load is calculated at 1.44gP/PE/day.

⁷ This is a licensed PTP discharge (NRW private communication), not yet appearing on the Rivers Trust mapping.

PE numbers are our estimates. We have used 3 for locations which look like individual dwellings⁸ and 10-20 for pubs and cafes. The numbers are an attempt to estimate time-averaged load, as opposed to peak load, or indeed, capacity. Bronllys Hospital is a special case. In 2019, Bronllys Hospital had 1 in-patient ward and a significant out-patient activity; we have called this 100 PE. Note that Table 17 is just a list of PTPs which are licensed and thus reported to the Rivers Trust. There are probably more.

Sub catchment	P load
Dulas Brook	0.20 kgP/day
River Llynfi above Dulas confluence	0.02 kgP/day
River Ennig	0.00 kgP/day
Full Llynfi catchment	0.42 kgP/day

Table 18: daily average phosphorus load from PTPs; 'P load' is calculated as 1.44gP/PE/day x PE

CSO Spills

Introduction

Combined storm overflows (CSOs) are a mechanism to protect sewer systems from overload and consequential backing up. In principle this should only occur in exceptionally heavy rainfall. However, due to chronic lack of investment to match local population growth and changing weather patterns, these 'safety valves' are increasingly discharging under modest provocation. Discharges can also be caused by blockages.

The majority of CSOs discharge untreated sewage via a basic 'screen', intended to filter out large items, but self-evidently a lot get through anyway. Those associated with storm tanks would claim to discharge partially settled effluent – meaning some reduction in faeces load. However, a reasonable approximation is that the contents of these discharges include the full input PE load of Table 8, albeit sometimes diluted by rainwater.

Discharges in 2020

According to the Rivers Trust data, there are 10 CSOs discharging into the Llynfi catchment. In 2020, there were 209 spills from 7 of these totalling 1273 hours:

Overflow Name	Monitored?	Discharges to	2020 Spills	2020 Hours
Trefecca Old Railway Maesgwyn	Y	Llynfi	39	513
Talgarth Along the Green	Y	Llynfi	12	12
Bronllys Bridge Manhole	Y	Llynfi	1	1
Talgarth STW	Y	Llynfi	44	198
Bronllys Rear of Garage A479	Y	Llynfi	68	297
Llangorse Pumping Station	?	Llynfi	0	0
Three Cocks Gwernyfred Park	Y	Felindre	8	21
Three Cocks Pumping Station (a)	Y	Tregoyd	37	231
Three Cocks	N	Felindre	0	0
Three Cocks Pumping Station (b)	N	Tregoyd	0	0
Bwlch North Pumping Station	N	Upper Llynfi	0	0
			209	1273

Table 19: known CSO discharges within the Llynfi catchment

⁸ The assumption here is that a typical dwelling is occupied by a couple with two children for 25 years, followed by 25 years without children – hence an average of 3; by contrast, a three-bedroom house would be expected to have a system capacity of 5 PE under current regulations.

All of these are located in the lower catchment around Talgarth, with the exception of Trefecca, Llangorse P.S. and Bwlch North P.S.

It seems highly unlikely that this is a complete list insofar as many overflows remain un-permitted. Indeed, we believe Llangorse P.S. (which we became aware of via an FoI response from DC-WW) to be one such. Note that the last three CSOs in the list had not yet been fitted with discharge monitoring, so the zeros simply signify an absence of data, as does the zero for Llangorse P.S.

At least four of these locations (Trefecca, Talgarth STW, Bronllys Garage & Three Cocks P.S.) are clearly in need of investigation by DC-WW.

Quantifying the discharges

Table 19 tells us only that a discharge occurred; there is no measure of volume. In order to get beyond this, we need to make a (very) rough estimate of how many PE are associated with each discharge. This works in two stages.

We first note that overflows are physically 'weir' systems, meaning that only a portion of the total load within the sewer is generally being discharged. In an attempt to be fair, we shall assume that this fraction is 25%; hence 75% of the load is still flowing to the STW downstream.

Next, we estimate how many PE are inputting to each discharge point. For STWs, this is simply the Input PE figure provided by DC-WW. For the CSO in Trefecca, we have used the local population figure. Around Talgarth and Bronllys, we have divided the Talgarth STW Input PE by the number of CSOs within its catchment. Similar at Three Cocks. All very approximate, but it should get us to the correct order of magnitude. So better than nothing.

Having thus defined the discharge fraction and the effective PE, we then calculate the emissions from each CSO using the Input PE definition of Table 8. The results are as follow:

Overflow Name	Effective PE	Liquid	Solids	BOD load	P load
Trefecca Old Railway Maesgwyn	456	439 m ³	609 kg	146.2 kgO ₂	6.1 kgP
Talgarth Along the Green	807	18 m ³	25 kg	6.1 kgO ₂	0.3 kgP
Bronllys Bridge Manhole	807	2 m ³	2 kg	0.5 kgO ₂	0.0 kgP
Talgarth STW	2,422	899 m ³	1,249 kg	299.7 kgO ₂	12.5 kgP
Bronllys Rear of Garage A479	807	450 m ³	624 kg	149.9 kgO ₂	6.2 kgP
Three Cocks Gwernyfred Park	74	3 m ³	4 kg	1.0 kgO ₂	0.0 kgP
Three Cocks Pumping Station (a)	74	32 m ³	44 kg	10.6 kgO ₂	0.4 kgP
		1,842 m³	2,558 kg	614 kgO₂	25.6 kgP

Table 20: estimated total pollution loads from CSO discharges within the Llynfi catchment in 2020; see Box 5 below for calculation details

Box 5: Calculation of CSO pollution loads

Taking 'Talgarth along the green' P load as an example:

CSOs in Talgarth & Bonllys:	3	excludes Talgarth STW
Input PE for Talgarth STW:	2,422	DC-WW data
Effective PE:	$= 2,422 / 3 = 807$	
Discharge fraction:	25%	our guess
2020 discharge duration:	12 hours = 0.5 days	from Table 19
Phosphorus load per PE:	2.5 g/PE/day	from Table 8
P load:	$= 2.5 \text{ gP/PE/day} \times 807 \text{ PE} \times 0.5 \text{ days} \times 25\%$ $= 0.252 \text{ kgP} \approx \mathbf{0.3 \text{ kgP}}$	

The thought of 2.6 tonnes of human excrement being dumped into the catchment is a sobering one. Beyond this, the totals in Table 20 show that CSO emissions are not very significant on an annual average basis for phosphorus and BOD. The primary problem here occurs during a discharge. Recalling that discharges can occur in summer as a result of a blockage or a heavy shower, we should also examine these discharges with respect to Q95 flows in the receiving water. This is shown below:

Overflow Name	Liquid dilution	BOD conc.	P conc.
Trefecca Old Railway Maesgwyn	0.4%	1.29 mgO ₂ /l	0.054 mgP/l
Talgarth Along the Green	0.3%	0.85 mgO ₂ /l	0.036 mgP/l
Bronllys Bridge Manhole	0.3%	0.85 mgO ₂ /l	0.036 mgP/l
Talgarth STW	0.8%	2.56 mgO ₂ /l	0.107 mgP/l
Bronllys Rear of Garage A479	0.3%	0.85 mgO ₂ /l	0.036 mgP/l
Three Cocks Gwernyfred Park	0.4%	1.44 mgO ₂ /l	0.060 mgP/l
Three Cocks Pumping Station (a)	0.4%	1.44 mgO ₂ /l	0.060 mgP/l

Table 21: estimated pollution loads during CSO discharges within the Llynfi catchment in 2020; see Box 6 below for details of calculation

Box 6: Calculation of CSO pollution concentrations

Taking 'Talgarth along the green' as an example:

Effective PE:	807	from Table 20
Discharge fraction:	25%	our guess
Liquid load per PE:	180 l/PE/day	from Table 8
CSO liquid flow:	$= 180 \text{ l/PE/day} \times 807 \text{ PE} \times 25\%$ $= 36,315 \text{ l/day} = 0.420 \text{ l/s}$	
Receiving water Q95 flow:	0.164 m ³ /s	from Table 2
Dilution:	$= \text{CSO liquid flow} / \text{Q95 flow} = \mathbf{0.3\%}$	
Phosphorus load per PE:	2.5 gP/PE/day	from Table 8
CSO P flow:	$= 2.5 \text{ gP/PE/day} \times 807 \text{ PE} \times 25\%$ $= 504 \text{ gP/day} = 5.84 \text{ mgP/s}$	
Receiving water P conc.	$= \text{CSO P flow} / \text{Q95 flow}$ $= (5.84 \text{ mgP/s}) / (0.164 \times 1000 \text{ l/s}) = \mathbf{0.036 \text{ mgP/l}}$	
... the calculation for BOD is analogous to that for P presented here		

Resulting BOD concentrations are all below the Water Framework Directive limit for *high* status of 3 mgO₂/l. However, the situation for phosphorus is more worrying, with every site taking its receiving water to excessive levels⁹. Recalling our earlier caveat that the assumptions underpinning this sequence of calculations are necessarily very approximate, we cannot draw precise conclusions here.

Nevertheless, it is reasonable to state that CSO emissions present a significant pollution event for the duration of a discharge. And, referring to Table 19, we can see that discharges can be extensive; Trefecca for example has an average duration of 13 hours. The average spill thus comprised 11 m³ of effluent, including 16 kg of solids, 3.7 kgO₂ of BOD and 0.15 kg of phosphorus. The Q95 flow rate at this point of the Llynfi is around 0.06 m³/s. It is hard to imagine that this is not a serious pollution event on such a small river.

Summing up phosphorus over our key sub catchments as before, we get:

Sub catchment	P load
Dulas Brook	0.000 kgP/day
River Llynfi above Dulas confluence	0.017 kgP/day
River Ennig	0.000 kgP/day
Lower Llynfi, Tregoyd & Felindre Brooks	0.053 kgP/day
Full Llynfi catchment	0.070 kgP/day

Table 22: daily average phosphorus load from CSOs

Unsurprisingly, we find that CSO pollution is centred around the major settlements in the lower catchment. Of particular concern are emissions directly from Talgarth STW due to its high PE load.

Box 7: Calculation of P load (Table 22)

For each sub catchment, we sum the annual CSO emissions falling within it (Table 20, 'P load'). We then divide by 365 to convert to a daily average value.

Taking 'River Llynfi above Dulas confluence' as an example, there is only one CSO load (Trefecca Old Railway) falling within it, therefore:

Total annual P load:	6.1 kgP	from Table 20
Daily average P load:	= 6.1 kgP / 365	
	= 0.017 kgP/day	

⁹ The calculations here refer to total phosphorus, therefore we cannot compare directly to the water quality target for RP.

Agriculture

On land in a near-natural condition the soil phosphorus content is relatively low, originating mainly from the mineralogy of the soil. Rivers draining such land thus carry little phosphorus – generally showing RP below 0.01 mgP/l. However, as we saw in the earlier section on legacy phosphorus, modern agriculture depends upon maintaining elevated levels of nutrients in the soil. It is axiomatic that this will increase the levels of phosphorus in the river. This is reflected in the fact that phosphorus targets for good ecological status in upland rivers, even where the other anthropogenic phosphorus sources are insignificant, have water quality targets for RP in the range 0.01-0.03 mgP/l.

There are multiple pathways by which agricultural phosphorus gets into a river. These are generally divided into two categories, diffuse and point source, which we shall consider separately.

Diffuse sources

Introduction

In dry conditions, water gets to the river by subsurface flow through the water table. But this water was originally rainfall and it got into the water table by percolating through the topsoil where it picked up a fraction of the prevailing phosphorus concentration (leaching). This phosphorus load may be modified by buffering processes in the subsoil en route to the river. When it rains, the water table rises, which switches on additional transfer routes - drains and ditches which were previously dry for example. These new routes are more direct and thus carry more of the topsoil phosphorus load.

Furthermore, during heavy rain we can also get surface run-off. This can further increase nutrient transfer by picking up recently applied fertiliser and washing it directly to surface water. It can also cause bulk soil erosion. However, this is primarily associated with the bare winter soils of arable land, where it is found to be the major nutrient transport route. It is considered less of a problem on permanent pasture.

Overall, agricultural phosphorus concentrations in the river have a tendency to increase with rainfall – and hence with river flowrate.

Export coefficients

The relationship between soil phosphorus concentration and leaching rates have been well studied at the small scale (*Heckrath et al., 1995; Smith et al. 1998*). However, extending such physically based models to the catchment level is fraught with difficulties. There are just too many unknowns. A cruder, but more generally accepted approach is that of *Burt et al. (1995)*. This uses area-based export coefficients to estimate nutrient loss from different land types and, separately, per capita nutrient losses from livestock. Essentially, these are lumped annual average numbers which, despite telling us nothing about seasonal variation or rainfall dependency, are a pragmatic way to build annual totals.

May et al. (2008), in their study of Llangorse Lake, used export coefficients derived from *Burt et al. (1995)*, considering these to be applicable to the Llangorse catchment. Since the Llangorse catchment is not untypical of the wider Llynfi catchment in terms of agriculture, topography and geology, we have used the same coefficients in this analysis. This also has the benefit of maintaining compatibility with this relevant prior work.

The export coefficients used are as follow:

Land use	Export Coef.
Arable	0.68 kgP/ha/y
Grazing	0.26 kgP/ha/y
Woodland/heath	0.02 kgP/ha/y
Livestock	
Cattle	0.22 kgP/ca/y
Sheep	0.03 kgP/ca/y

Table 23: phosphorus export coefficients for land and livestock, from May et al. (2008); note that land use coefficients are per hectare and livestock coefficients are per capita (head)

Thus, the *Land use* export coefficients in Table 23 represent the phosphorus load from the land, arising simply as a consequence of the phosphorus concentration and other relevant conditions typically encountered on such land. Note that the *Grazing* export coefficient is an order of magnitude larger than that for *Woodland /heath*, the latter having only natural levels of soil phosphorus. When we put livestock onto grassland, the concomitant manure phosphorus load is additional, and further increases the phosphorus export pro-rata to the stock density. The *Livestock* export coefficients reflect this contribution.

According to the current CPRW data, there are no intensive poultry units within the Llynfi catchment to complicate the calculation. The presence of intensive pig units however is unknown¹⁰.

We can now take the catchment land cover data presented in Table 3, plus the livestock estimates from Table 4, and calculate an annual phosphorus export as follows:

Source	Quantity	P load
Arable	1,247 ha	848 kgP/yr
Grazing	9,540 ha	2,480 kgP/yr
Woodland	1554 ha	31 kgP/yr
Heath/bog	360 ha	7 kgP/yr
Sheep	85,000 ca	2,557 kgP/yr
Cattle	4,000 ca	864 kgP/yr
		6,788 kgP/yr

Table 24: P export from diffuse agricultural sources; Quantity is from Tables 3 & 4; P load is Quantity x Export Coef. (Table 23)

Stated as a daily average by sub catchment, we have:

Sub catchment	Area fraction	P load
Dulas Brook	30%	5.63 kgP/day
River Llynfi above Dulas confluence	45%	8.34 kgP/day
River Ennig	17%	3.10 kgP/day
Full Llynfi catchment	100%	18.6 kgP/day

Table 25: daily average phosphorus load from diffuse agricultural sources; Area fraction is derived from Table 2; P load is 6,788 kgP/yr x Area fraction / 365

¹⁰ It is thought that many pig units do not have explicit planning consent, either because consent was never sought, or because the application sneaked through as a 'livestock shed' or similar, thus concealing the environmental implications. We are aware of at least one pig unit, adjacent to the Ellywe, a tributary of the Ennig. This is suspected to have been responsible for significant levels of pollution resulting from uncontrolled surface run-off. We understand that the situation has been addressed following NRW enforcement action. But the point is that there are potentially many more, which we are currently unaware of.

It is worth pointing out that the above numbers represent a best-case situation, being applicable to land over 50m from a watercourse. Below 50m, the livestock coefficients double and the land coefficients also increase (*Burt et al. 1995*). Effectively this means that the numbers used here assume strict adherence to COGAP manure spreading guidance and livestock fenced away from surface water. Almost certainly not the case universally.

Point sources

Point source pollution has been reviewed by *Edwards*. The term encompasses a wide variety of sources, most of which are almost impossible to estimate across a catchment. Agricultural point sources would include run off from farmyards and other hard standings, run off from farm tracks and poached ground by gates and feeding stations, leaching from slurry pits, poorly located manure heaps, livestock accessing the river. The list goes on.

Rather than attempt to put a number to this category, I will simply note that it could easily be very significant. *May et al. 2008* referred to some point source categories as potential explanations for their source apportion model for Llangorse predicting lower lake P levels than were observed.

However, we should note that the data used by *Burt et al. (1995)* to calculate their export coefficients was collected in a real-world farming environment. As such, it is reasonable to conclude that these coefficients unavoidably include a broad selection of common 'point source' pollution types. For example, poaching due to high stock densities was specifically discussed by these authors.

Total Load (source apportionment model)

We can now sum up the different phosphorus loads for each sub catchment. We can also divide the loads by the corresponding river mean flowrates to determine mean phosphorus concentrations (as total phosphorus). Firstly, we will look at the three main tributaries separately, then we will look at the full Llynfi catchment.

Dulas Brook

For this we will use our estimate that the mean flowrate (Q_m) for the Dulas Brook is $0.7 \text{ m}^3/\text{s}$ (Table 2). In the following table, we sum the inputs from anthropogenic phosphorus sources estimated earlier and divide by the mean flowrate to estimate the resulting mean phosphorus concentration in the river.

Source	P load	Fraction	$P_{\text{conc}} @ Q_m$
STW	0.1 kgP/day	1%	0.001 mgP/l
PTP	0.2 kgP/day	3%	0.003 mgP/l
CSO	0.0 kgP/day	0%	0.000 mgP/l
Septic tanks	0.7 kgP/day	10%	0.011 mgP/l
Diffuse agriculture	5.6 kgP/day	86%	0.094 mgP/l
Total	6.6 kgP/day	100%	0.109 mgP/l

Table 26: total phosphorus loads on the Dulas sub-catchment; P load is from Tables 12,16,18,22,25; $P_{\text{conc}} @ Q_m$ is the estimated mean river P concentration at mean flowrate, calculated as P load (mgP/s) / Dulas mean flowrate (700 l/s) from Table 2

Looking at this graphically:

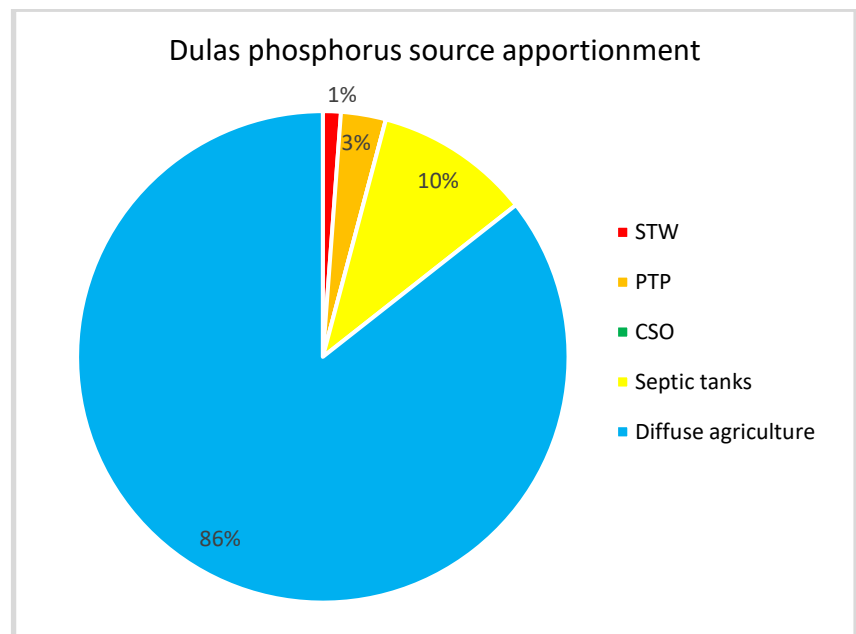


Figure 5: phosphorus source apportionment for the Dulas sub-catchment

On an annual average basis (average load and average flowrate), the Dulas appears to be heavily agriculture dominated (86% of load), with septic tanks the next largest source (10%). The STW and PTP load is minor by comparison and, as we saw in Table 13, the STW effluent is probably still reasonably dilute at Q95 flows. The estimated mean flowrate phosphorus concentration ($P_{\text{conc}} @ Q_m$) of 0.109 mgP/l is alarmingly high.

River Llynfi above the Dulas confluence

Our estimate for the mean flowrate of this section of river is 1.03 m³/s (Table 2). Using the same procedure as for the Dulas, the source apportionments are as follow:

Source	P load	Fraction	P _{conc} @ Q _m
STW	0.0 kgP/day	0%	0.000 mgP/l
PTP	0.0 kgP/day	0%	0.000 mgP/l
CSO	0.0 kgP/day	0%	0.000 mgP/l
Septic tanks	1.0 kgP/day	11%	0.011 mgP/l
Diffuse agriculture	8.3 kgP/day	89%	0.094 mgP/l
Total	9.3 kgP/day	100%	0.105 mgP/l

Table 27: total phosphorus loads on the Llynfi sub-catchment, above the Dulas confluence; P load is from Tables 12,16,18,22,25; P_{conc} @ Q_m is the estimated mean river P concentration at mean flowrate, calculated as P load (mgP/s) / Llynfi mean flowrate (1030 l/s) at this point, from Table 2

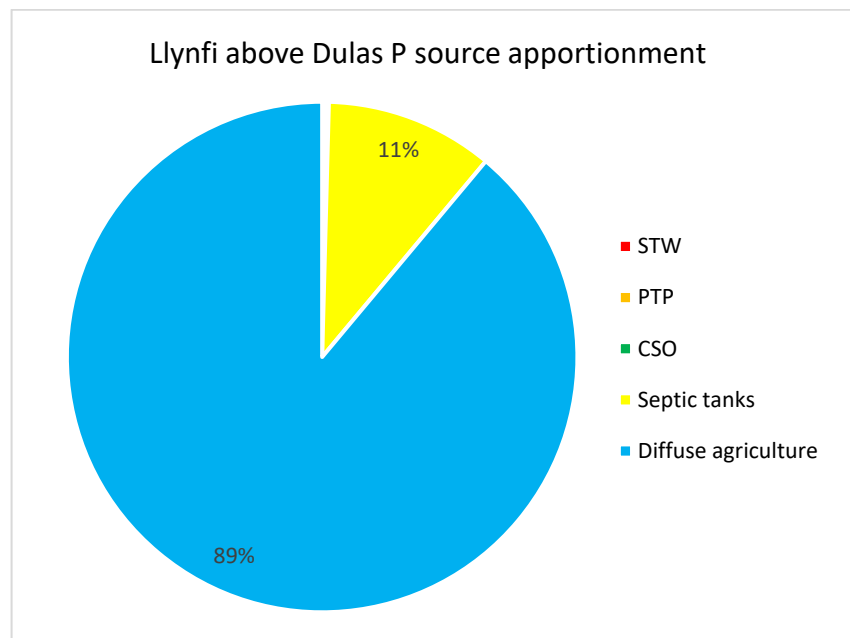


Figure 6: phosphorus source apportionment for the Llynfi sub-catchment, above the Dulas confluence

There are no STWs or PTPs within the Llynfi sub-catchment, above the Dulas confluence. Hence, we see total domination by agriculture (89%), followed by septic tanks (11%). The estimated mean flowrate phosphorus concentration (P_{conc} @ Q_m) of 0.105 mgP/l is, once again, alarmingly high.

River Ennig

Our estimate for the mean flowrate of this section of river is $0.38 \text{ m}^3/\text{s}$ (Table 2). Using the same procedure as for the Dulas, the source apportionments are as follow:

Source	P load	Fraction	$P_{\text{conc}} @ Q_m$
STW	0.0 kg/day	0%	0.000 mgP/l
PTP	0.0 kg/day	0%	0.000 mgP/l
CSO	0.0 kg/day	0%	0.000 mgP/l
Septic tanks	0.4 kg/day	11%	0.011 mgP/l
Diffuse agriculture	3.1 kg/day	89%	0.094 mgP/l
Total	3.5 kg/day	100%	0.105 mgP/l

Table 28: total phosphorus loads on the Ennig sub-catchment; P load is from Tables 12,16,18,22,25; $P_{\text{conc}} @ Q_m$ is the estimated mean river P concentration at mean flowrate, calculated as P load (mgP/s) / Ennig mean flowrate (380 l/s) from Table 2

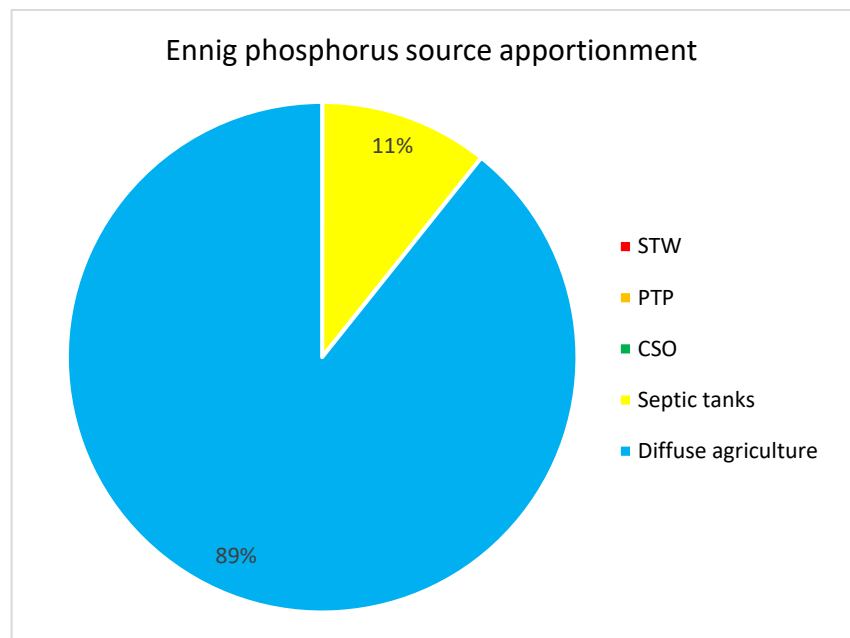


Figure 7: phosphorus source apportionment for the Ennig sub-catchment

The Ennig sub-catchment does not contain any STWs or PTPs. There is one CSO in the centre of Talgarth, but this didn't discharge much in 2020. Consequently, we see only diffuse agriculture (89%) and septic tanks (11%) featuring in the pie chart of Figure 7.

Full Llynfi catchment

Using the measured flowrates at Three Cocks (Table 2) the full catchment source apportionments are as shown in Table 29. Because we have measured flowrates at this point, rather than estimates, we also include results for phosphorus concentration at Q95 flowrate for the three 'constant mass flow' phosphorus sources.

Source	P load	Fraction	P _{conc} @ Q _m	P _{conc} @ Q95
STW	1.6 kgP/day	7%	0.008 mgP/l	0.113 mgP/l
PTP	0.4 kgP/day	2%	0.002 mgP/l	0.027 mgP/l
CSO	0.0 kgP/day	0%	0.000 mgP/l	0.005 mgP/l
Septic tanks	2.2 kgP/day	10%	0.011 mgP/l	
Diffuse agriculture	18.6 kgP/day	81%	0.094 mgP/l	
Total	22.9 kgP/day	100%	0.115 mgP/l	

Table 29: Total phosphorus loads for the full Llynfi catchment; P load is from Tables 12,16,18,22,25; P_{conc} @ Q_m is the estimated mean river P concentration at mean flowrate, calculated as P load (mgP/s) / Llynfi mean flowrate (2300 l/s) from Table 2; P_{conc} @ Q95 uses the Q95 flowrate (160 l/s)

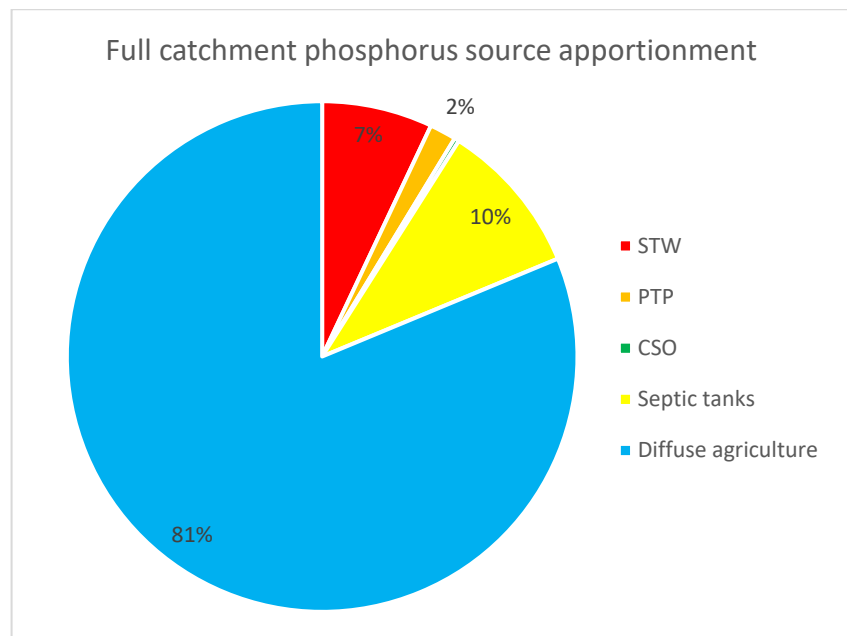


Figure 8: P loads for the full Llynfi catchment (kg/day)

We can see from the total for the P_{conc} @ Q_m column in Table 29, that the Lower Llynfi carries an extremely high total phosphorus load. We shall defer to the next section for consideration of what that implies for river RP measurements.

In the lower reaches, the river has picked up additional input from STWs and PTPs, taking their combined total to 9%. Nevertheless, diffuse agriculture (81%) still dominates the picture when the river is at mean flowrate. But the P_{conc} @ Q95 column points to another problem. At Q95 flowrate, the reduced dilution of STW and PTP effluent increases their combined contribution to river phosphorus concentration from 0.01 mgP/l to 0.14 mgP/l. Thus, the lower river transitions from being agriculture dominated at mean flow to being sewage dominated in the lower flows of summer.

Discussion

At this point, it is useful to summarise and review what the source apportionment modelling is telling us. We must acknowledge the approximations and assumptions made in each of the five pollution categories which we have attempted to quantify. Export coefficients can only ever be an approximation to a highly complex reality. Furthermore, some of the head counts and populations are occasionally rough estimates. That said, some conclusions can be stated with confidence.

Upper sub catchment

The catchment above the Ennig confluence is rural, with a low population density. Consequently, each of the sub-catchments analysed, the Dulas Brook, the Llynfi and the Ennig all carry heavy phosphorus loads, with diffuse agriculture as the dominant source making up 86% - 89% of the total load. Septic tanks feature next, at 9% to 11%. The only STW within the upper catchment is Llanfilo on the Dulas Brook. This has a very small load, which is unlikely to have serious environmental impact in normal operation. Other sources are minor or non-existent.

Lower sub catchment

Below the Ennig confluence, the Llynfi picks up more sewage load from the three STWs in the lower sub catchment. This results in STWs contributing 7% of the total phosphorus load. PTPs also feature more, contributing 2% of total load.

Comparison to Q95 flowrates show that Talgarth STW, Felindre STW and Aberllynfi STW suffer from poor dilution in their respective receiving waters during summer. Talgarth STW was much improved by the installation of P-stripping in 2015 but remains a concern at summer flow. The proposal to further enhance this STW in the AMP8 investment cycle is to be welcomed. The proposal to install P-stripping at Aberllynfi is likewise to be welcomed. The combined effect will be to halve the STW phosphorus load within the catchment. However, Felindre remains to be addressed.

River phosphorus levels and diffuse sources

Throughout the Llynfi catchment, river concentrations of total phosphorus are estimated to be in the region of 0.11 mgP/l, resulting entirely from diffuse agricultural and septic tanks. Below Talgarth, the load is increased somewhat by STW effluent. But the principal problem to solve in the Llynfi catchment is clearly diffuse pollution. Moreover, the predicted river phosphorus levels are around three times where they need to be to have any chance of meeting SAC EQS targets for reactive phosphorus.

An interesting observation drops out of this simple modelling, which may not be so obvious in more sophisticated modelling schemes. The observant reader may have noticed that the same numbers crop up for the predicted river phosphorus concentrations due to diffuse agriculture and septic tanks in each of the sub-catchments (Tables 26, 27, 28). That is, diffuse agriculture and septic tanks together result in an average annual total phosphorus concentration of 0.11 mgP/l. This is not a coincidence. It arises because both the effective rainfall (hence the mean river flowrate) and the diffuse phosphorus loads are calculated on an area basis. So, the ratio becomes a fixed value, irrespective of the area we calculate over. Stated mathematically, we have:

$$\text{River P concentration} = \frac{\sum \text{Diffuse loads per area}}{\text{Effective rainfall per area}}$$

Since nothing can be done about rainfall, if river P concentrations are to be reduced, we need next to look at the diffuse loads on a per area basis. This is shown in the following chart:

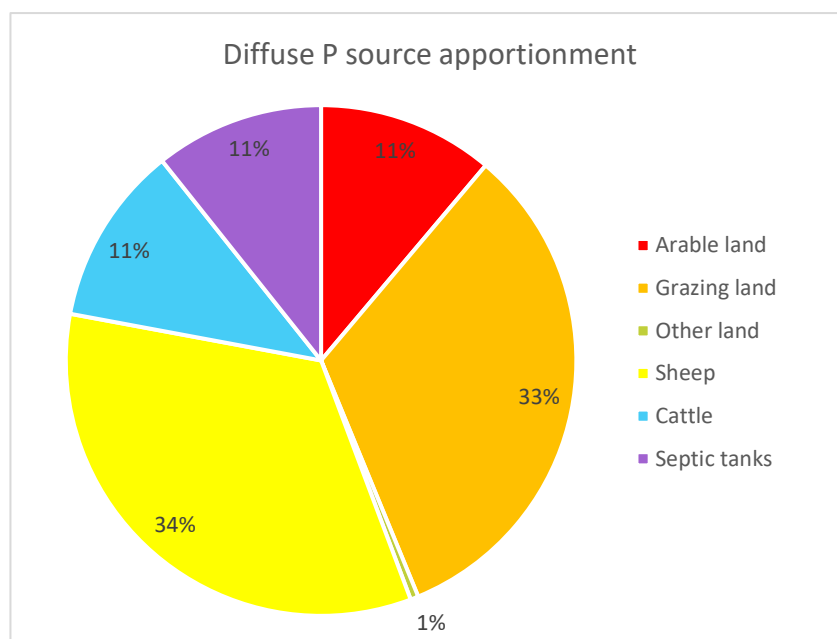


Figure 9: diffuse phosphorus sources calculated on a per area basis

One third of the total diffuse pollution comes purely from the presence of (fertilised) grazing land, before we even put livestock on it. A further third then comes from the stock density of sheep alone.

The inescapable conclusion from Figure 9 is that, if the objective is to reduce phosphorus levels by a factor of three, then it is necessary to substantially reduce the density of livestock and re-wild much of the pasture. Anything less than this simply will not resolve the problem.

Before leaving this topic, an interesting way to think of this problem is to consider a hectare of sheep pasture. After all, most of the catchment is pasture and most of that pasture has sheep on it. So, in order to fix the catchment, it is necessary to fix sheep pasture. Box 8 shows the calculation.

Box 8: Calculation of P export from 1 hectare of sheep pasture

Export coefficient for grassland:	0.26 kgP/ha/year	from Table 23
Export coefficient for sheep:	0.03 kgP/ca/year	from Table 23
Sheep stock density:	9 ca/Ha	from Table 4
Resulting sheep export:	$= 9 \times 0.03 = 0.27 \text{ kg/ha/year}$	
Total export (land + sheep):	$= 0.26 + 0.27 = 0.53 \text{ kgP/ha/year}$	
Hydrologically effective rainfall:	5,495 m ³ /ha/year	Llynfi mean flow / area
Resulting P concentration:	$= \text{Total export} / \text{HER}$	
	$(0.53 \times 10^6) / (5,495 \times 10^3) = \mathbf{0.10 \text{ mgP/l}}$	

This simple calculation illustrates the core reason why the Llynfi massively exceeds SAC EQS limits. Too much phosphorus in the soil and too many sheep, in equal measure. All else follows from this.

By way of comparison, consider the situation pertaining in the early part of the 20th century. This pre-dates the widespread use of synthetic fertilisers, at a time when Welsh sheep number were in the 3-4M region (cf. 9M today). Pasture phosphorus levels were presumably close to natural. Running the above calculation for 3.5 sheep per hectare and a land export coefficient of 0.02 kgP/ha/year reduces the resultant P concentration to 0.023 mgP/l – probably low enough to meet current RP targets.

Validation

In order to validate the source apportionment model developed above, we need to compare the results to water quality measurements. Before proceeding, there are two key aspects of measurement to clarify.

Phosphorus units

Thus far we have been estimating total phosphorus (TP) as phosphorus. That is, the mass of phosphorus within an emission, irrespective of its chemical form. Hence units such as mgP and kgP refer to mass of phosphorus. Likewise, concentration is stated as mgP/l, this being the mass of phosphorus per litre of water. Such measurement ‘as phosphorus’ is common in water quality reporting and is how EQS standards are specified.

However, hand-held instruments commonly measure and display ‘as phosphate’ – that is, the mass of the phosphate ion, PO_4 per unit volume of water. Specifically, the Hanna Checker, used by WSA in the work reported below, is one such. It displays ppm PO_4 , which is the same as mgPO_4/l , since one litre of water weighs 10^6 mg. The mass of the phosphate ion is 3.06 times the mass of the phosphorus atom within it (i.e., 3.06 mgPO_4 is the same as 1 mgP). Therefore, all Hannah Checker results below have been divided by 3.06 to bring them to consistent ‘as phosphorus’ units.

Phosphorus fractions

Phosphorus is present in a wide variety of forms in natural waters, in both dissolved and particulate phases, and in inorganic and organic forms, including biomass. Moreover, these forms are highly inter-reactive. Practical measurement techniques as discussed by *Jarvie et al. (2002)*, measure fractions of phosphorus, which do not conform neatly to specific forms of phosphorus.

The WFD (Standards and Classification) Directions (England and Wales) 2015 specifies phosphorus in terms of “reactive phosphorus” (RP), which is defined as “the concentration of phosphorus as determined using the phosphomolybdenum blue colorimetric method”, using filtration to $0.45\mu\text{m}$ “where necessary to ensure accuracy”. NRW do not use filtration, preferring to settle and decant samples (*Hatton-Ellis & Jones, 2021*); they report compliance data as “orthophosphate as P”.

WSA citizen science (CS) teams use hand-held instruments (Hanna Phosphate Checkers) which use the phosphomolybdenum blue method, but without filtration. Therefore, some particulate forms of phosphorus will be included in the WSA’s Hanna Phosphate Checker assays. A [recent report](#) by Cardiff University, analysing CS data from across the entire Wye catchment, has shown excellent cross correlation for RP between Hanna readings and laboratory analysis using unfiltered samples.

In order to compare our model predictions with measurement data, we thus need to take a view on what fraction of total phosphorus is represented by phosphomolybdenum blue assay. It has been established in the literature that different effluent sources show distinctive RP fractions. Sewage and livestock faeces are both high in RP; agricultural run-off is generally lower (for example, *Ekholm and Krogerus, 2003*). EA data from the Wye catchment over the period 1991-2000 was analysed by *Jarvie et al. (2003)*. This dataset predominantly predated the installation of phosphorus stripping at any STWs, and the catchment below Erwood was characterised by the authors as STW-dominated, with mean RP at 86% of TP. At the other end of the scale, we have recent sonde data from WUF showing a mean RP at 30% of TP on the agriculture-dominated Ithon (the principal STW on the Ithon has P-stripping in place). *May et al. (2008)* measured RP at 60% of TP on the Llynfi outflow from Llangorse Lake. The CS dataset analysed by Cardiff University in 2021 showed RP at approximately 80% of TP at Pipton Bridge on the lower Llynfi, albeit a single data point. Measurements made by NRW on the Llynfi above the Wye

confluence during 2018-19 (*Fol request Feb-22*) show an average ratio of 62.4%. For the following comparison we will take a central value and assume that the RP/TP fraction is 65%.

Predicted RP concentrations

Based on this RP/TP ratio, we can calculate the predictions of our source apportionment model in terms of RP:

Location	TP	RP
Dulas Brook	0.109 mgP/l	0.071 mgP/l
Llynfi above Dulas conf.	0.105 mgP/l	0.069 mgP/l
River Ennig	0.105 mgP/l	0.068 mgP/l
Llynfi (lower)	0.115 mgP/l	0.075 mgP/l

Table 30: predicted RP concentrations; TP is mean total P concentration (P_{conc} @ Q_m) from Tables 26, 27, 28, 29; RP is calculated as 65% of TP

For comparison, we have available four sets of data as follows.

Countryside Council for Wales (CCW) 2008 & NRW 2018

May et al. (2008) reported measurements immediately below the Llangorse outflow to the Llynfi. Annual mean RP was 0.08 mgP/l.

The later study by *Goldsmith et al. (2018)* reported TP results (but not RP) for the period 2014-18. TP levels over this period were found to be similar to the 2008 value.

Thus, the available evidence for Llangorse is in good agreement with our model.

Natural Resources Wales (NRW) 2017-19

This compliance dataset was reported by *Hatton-Ellis & Jones (2021)*. Results for the Llynfi catchment are summarised as follows:

Waterbody ID	Waterbody name	EQS target	Samples	Annual mean
109055036970	Triffwrdd – src. To Dulas conf.	0.015 mgP/l	14	0.070 mgP/l
109055036920	Dulas Bk. – src. To Llynfi conf.	0.025 mgP/l	9	0.074 mgP/l
	Llynfi above Dulas conf.	Not reported		
	River Ennig	Not reported		
109055036920	Llynfi (lower)	0.025 mgP/l	26	0.077 mgP/l

Table 31: NRW phosphorus compliance data for 2017-19: EQS target & Annual mean are RP

No data is provided for the Llynfi above the Dulas confluence, for the River Ennig, or for Llangorse Lake. The results for the Dulas, its tributary the Triffwrdd and Lower Llynfi are in good agreement with the source apportionment model presented here, albeit with a fairly low sample count.

Note that all measured values (*Annual mean* in Table 31) are 3-4x the EQS target for the waterbodies. This is not a marginal failure.

Natural Resources Wales 2020-21

Some more recent NRW data has been obtained via a freedom of information (Fol) request. Unfortunately, NRW did very little monitoring during 2020 due to Covid restrictions so, in

practice, this dataset relates primarily to 2021. But it fills in an important gap on the Middle Llynfi (above the Dulas confluence).

Station No.	Station name	Samples	Mean RP
S50091	Llynfi, Llangorse Lake Outlet	3	0.038 mgP/l
S50092	Llynfi at Llynfi Bridge Llangorse	4	0.056 mgP/l
S50093	Llynfi above Dulas Brook	11	0.099 mgP/l
S50098	Llynfi at conf. with R. Wye	13	0.064 mgP/l

Table 32: NRW phosphorus compliance data for 2020-21

Wye Salmon Association (WSA) 2020-21

This dataset is the result of a citizen science (CS) project carried out by The Wye Salmon Association. Results for the Llynfi catchment are summarised as follows:

Waterbody name	Samples	Annual mean RP
Llynfi (lower) at Pipton Bridge	51	0.047 mgP/l
Llynfi at Llynfi Bridge Llangorse	18	0.077 mgP/l
Total dataset (11 locations)	116	0.049 mgP/l

Table 33: WSA RP data for July 2020- August 2021

Pipton Bridge is on the lower river, near Three Cocks, and importantly is below Talgarth STW. The annual mean value here is slightly lower than the NRW value for the lower Llynfi. But note that the NRW number is based on far fewer samples.

Llynfi Bridge is 1.1km below the outflow from Llangorse Lake. The value here is consistent with the result of *May et al. (2008)*, whose measurements were taken just 100m below the outflow – hence, 1km upstream of this point.

The above data are summarised in the following table:

Waterbody name	SAM	CCW	NRW (a)	NRW (b)	WSA
Llynfi (lower)	0.075 mgP/l		0.077 mgP/l	0.064 mgP/l	0.047 mgP/l
Dulas Brook	0.071 mgP/l		0.070 mgP/l		
Tryffrwd Brook			0.074 mgP/l		
Ennig	0.068 mgP/l				
Llynfi above Dulas con.	0.069 mgP/l	0.080 mgP/l		0.099 mgP/l	0.077 mgP/l

Table 34: summary of validation data; SAM is our source apportionment model prediction; CCW is from *May et al. (2008)*; NRW(a) is 2017-19 data from *Hatton-Ellis et al.(2021)*; NRW(b) is 2020-21 data obtained via Fol request; WSA is 2020-21 Wye Salmon Association data

Overall, we see remarkably good agreement between our simple source apportionment model and measured values of annual mean RP on the river, based on a reasonable assumption about the RP fraction. The model is obviously not far off the mark. And we haven't need to top up with any 'point source' inputs. Everything is explained by business as usual across the catchment.

Load apportionment modelling

Jarvie *et al.* (2003) observed that the Llynfi had a broadly negative relationship between RP concentration and flow, meaning that concentration was strongly influenced by point source (STW) loads being diluted in higher flow conditions. Plotting the WSA dataset as concentration versus volumetric flowrate does indeed show a negative relationship at both Pipton Bridge and Llangorse Bridge. Moreover, the graphs are qualitatively similar to the one presented by Jarvie *et al.*

However, we can go further with this analysis. Bowes *et al.* (2008) developed a mathematical model to describe the concentration-flow relationship, as follows:

$$C_p = A \cdot Q^{B-1} + C \cdot Q^{D-1}$$

where C_p is river phosphorus concentration, Q is volumetric flow rate and A , B , C & D are load coefficients to be determined empirically.

Coefficient B relates to instream (silt) absorption and release processes, which are not usually significant in eutrophic rivers. So, we can set $B=0$ and our function then simplifies to:

$$C_p = A \cdot Q^{-1} + C \cdot Q^{D-1}$$

The first term is the nutrient¹¹ concentration originating from fixed mass flow sources, primarily point sources such as STWs, PTPs and 'misconnected' septic tanks. The second term is the nutrient concentration originating from rainfall and flow-related sources and will largely equate to diffuse sources derived from agriculture, road run-off and septic tank soakaways.

The model can readily be programmed into Microsoft Excel and the parameters A , C & D optimised by the Excel Solver function to minimise residual variance in the dataset. This process is referred to as load apportionment modelling (LAM). Since 2008, it has been successfully applied to various catchments, including the Dorset Frome (Bowes *et al.*, 2009), the Thames Basin (Bowes *et al.* 2014) and more recently the Great Lakes (Rattan, Bowes, Yates, Culp & Chambers 2021).

Pipton Bridge

The plot of the 51 measurements taken by WSA at Pipton bridge is shown in Figure 10 below. The red curve is the optimised LAM function. This shows a marked negative slope at low flows, transitioning into a slight positive slope at higher flows. The model accounts for 14% of the dataset variance, which is significant at 90% confidence.

The transition point, the point where the two terms in the model are equal (Q_e), can be calculated from the optimised coefficients using the following relationship:

$$Q_e = \left(\frac{A}{C}\right)^{(1/(D-B))}$$

For this dataset, $Q_e = 0.20 \text{ m}^3/\text{s}$. This flow rate corresponds approximately to the 80% exceedance level referenced back to the Three Cocks gauge. What this indicates is that for approximately ten weeks during the summer, when the river is low, the phosphorus load is dominated by point source sewage effluent. For the remainder of the time, the load is dominated by diffuse sources.

We can extract another useful piece of information from this model. The A coefficient equates to the mass flowrate of nutrient arising from point source sewage effluent. For the Pipton dataset, we have:

$$A = 0.0061(\text{mg} \cdot \text{l}^{-1})(\text{m}^3 \cdot \text{s}^{-1})$$

¹¹ Although we are concerned here with phosphorus, the analysis method can be applied to other nutrients

Converting this to more friendly units tells us that the RP flow rate is 0.53 kg/day. We then convert this to total phosphorus, once again using our rough estimate of $RP/TP = 65\%$. This suggests that the observed RP signal results from a total phosphorus emission 0.82 kg/day. Now, immediately upstream of Pipton Bridge, we have Talgarth STW and Felindre STW; Table 11 tells us that the estimated emissions from these are 0.87 kgP/day and 0.23 kgP/day respectively. But Talgarth has phosphorus stripping, and its figure represents a maximum limit, not an operational target. If we speculate that DC-WW aim for say 70% of the limit, in order to provide headroom, then the total for these two STWs is equal to 0.84 kgP/day. So, our LAM estimate is falling well within $\pm 10\%$ of the expected value for STW emissions.

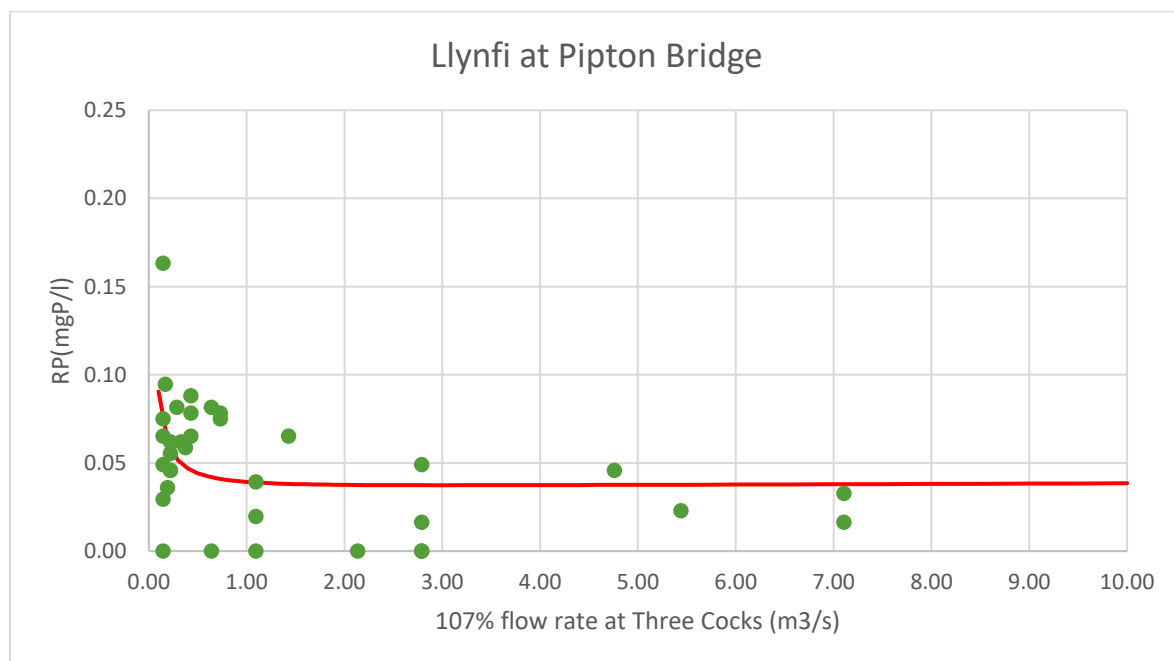


Figure 10: scatter graph of data from Pipton Bridge, with the optimised LAM function; the Felindre Brook confluence is above Pipton Bridge, but below the Three Cocks gauging station; its catchment is estimated at 10 km² (Table 2), 7% of the Three Cocks catchment, hence flowrates at Pipton Bridge are estimated to be 107% of Three Cocks flowrates.

Llynfi Bridge Llangorse

WSA have taken 24 measurements at Llynfi Bridge, located 1.1km below the Llangorse outfall. As discussed earlier, sewage from Llangorse Village is routed to Talgarth, so we would not expect to see a point source sewage signal, like the one observed below Talgarth.

There is no flow gauging on the upper river, therefore flowrates have been derived from the Three Cocks gauging downstream, by taking the estimated fraction of catchment area above Llynfi Bridge (17%). Given that the overall Llynfi catchment is small, there are unlikely to be major variations in weather patterns across it. Furthermore, the flow time between Llangorse and Three Cocks is just a few hours, thus limiting timing errors. Nevertheless, the presence of the lake will reduce the flashiness of this section of river relative to Three Cocks. Thus, we would expect some resulting increase in data variance. The plot is shown in Figure 11 below.

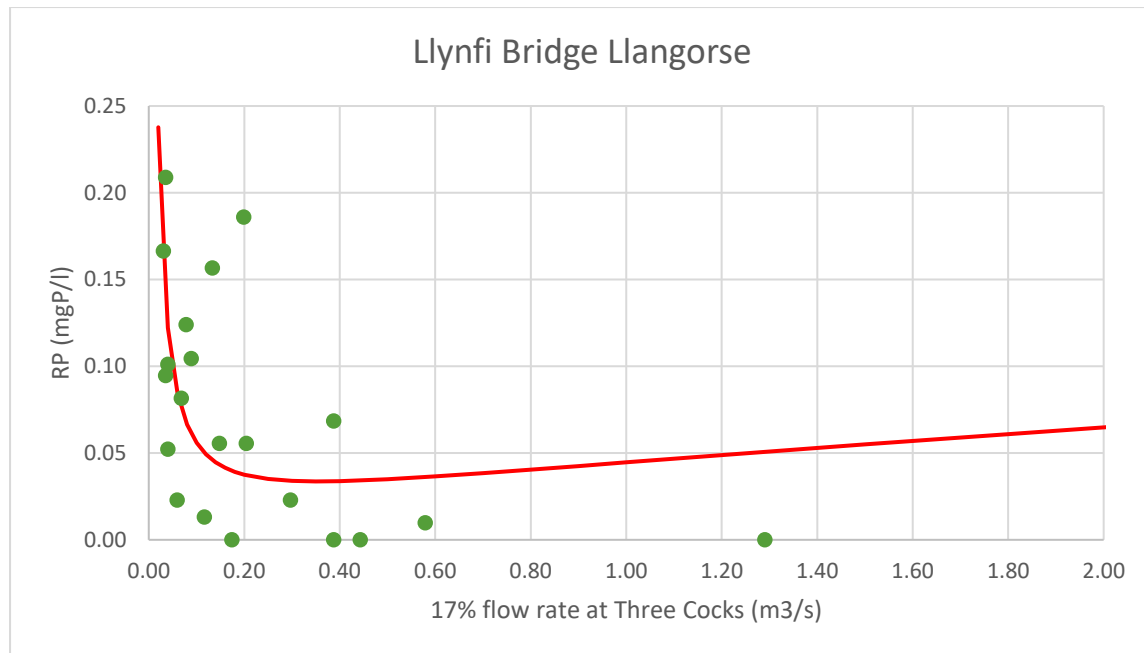


Figure 11: scatter graph of data from Llynfi Bridge Llangorse, with the optimised LAM function; catchment above Llynfi Bridge is 17% of the Three Cocks catchment; hence flowrates at Llangorse Bridge are estimated to be 17% of Three Cocks flowrates.

As before, the red curve shows the optimised LAM function. There is once again a strong negative slope region on the left side of the graph, indicative of a strong point source sewage influence. It seems unlikely that Llangorse Lake would exhibit such a signal. It is much more likely to be the result of a discharge directly into the river upstream of the measuring point. There is also a more pronounced flow dependence from the diffuse component. This model accounts for 31% of the dataset variance and, again, is significant at 90% confidence.

For this dataset, we have $Q_e = 0.27 \text{ m}^3/\text{s}$. Referencing this back to the Three Cocks gauge, this corresponds to the 45% exceedance value. Thus, this suggests that this section of the Llynfi spends approximately half of the time in a sewage dominated condition.

This is somewhat unexpected. We are aware of no point sources which would explain this signal. Using the A parameter, as we did for the lower Llynfi above, we can estimate the point source load being carried at Llynfi Bridge as 0.62 kgP/day (TP). From our source apportionment model, the estimated annual average diffuse loads at this point total 3.6 kgP/day. So, this unidentified, fixed-flow source is substantial. It equates to the treated effluent from around 400 people, or perhaps the regular wash-off from a milking parlour.

Within the WSA dataset examined, there are five occasions when readings were taken on the same day at the Car Park Footbridge, approximately 900m upstream of Llynfi Bridge, and at Llynfi Bridge; all of these were in low, summer conditions (July-August). The mean RP concentration at the footbridge from this data was 0.03 mgP/l, compared to 0.37 mgP/l at Llynfi Bridge. This strongly supports the conclusion that there is a significant point source of pollution within the 900m between the two bridges.

The source of this pollution needs to be investigated. There are a few obvious candidates:

- The CSO at Llangorse Pumping Station is located immediately upstream of Llynfi Bridge. As discussed above, it does not appear on Rivers Trust mapping and thus has no discharge

monitoring data reported. But certainly, has the theoretical capacity to output this quantity of phosphorus, handling sewage from Llangorse Village with a resident population of around 350 people.

- Also of particular concern is the adjacent caravan park. As discussed earlier, there are well over 200 fixed caravans & chalets on this site. It is currently unclear to WSA what the sewage arrangements are for this large seasonal population.
- In addition, the Tawel Brook plus a couple of un-named ditches join the Llynfi from the west. Any of these could be carrying an agricultural or 'misconnected' septic tank load.

The River Wye SAC nutrient management plan

The following is an abstract from the current Nutrient Management Plan for the River Wye SAC. It is included to provide background to our conclusions and recommendations.

Phosphate Action Plan, September 2021

Three main types of monitoring are conducted by NRW:

- Surveillance: designed to assess general environmental trends or patterns, with no specific targets. Surveillance sites are often monitored long-term and at higher frequency than other types of monitoring
- Monitoring: where a series of measurements are taken with the aim of comparing them to a specific target. This is used for compliance reporting and to identify whether a problem exists
- Investigations: these are generally time-limited pieces of work designed to diagnose the causes of failures detected by monitoring or by incident reports, and to identify a suitable course of action

NRW monitored phosphorus at more than 50 sample points in the Wye and its tributaries in Wales and these provided an overall picture of phosphorus concentrations in sub-catchments (*Hatton-Ellis et al., 2021*). All the middle Wye tributaries, including the three water bodies in the Llynfi sub-catchment, failed to meet their SAC RP targets.

The River Wye SAC Nutrient Management Plan (NMP) proposes that phosphate pollution be addressed on a sub-catchment basis and all sub-catchment plans include the same generic actions:

- Targeted work based on phosphates evidence report but also taking into account other relevant evidence
- Identify potential problem point sources and loading discharging to the river network including DCWW, private works, septic tanks, Combined Sewer Overflows, agricultural units, anaerobic digester plants
- Identify sources discharging to ground including septic tanks, land spreading etc.
- Identify and tackle pathways of runoff from land
- Where large changes to farming practice are occurring, use catchment officers to approach farmers and encourage changes to farm design, including much wider buffer strips, interception ponds to trap runoff and slow flow etc.
- Need for river restoration work along main river and tributaries to reduce pollution risk and ecological resilience
- Ditch blocking and wetland restoration to increase baseflows
- Encourage natural flood management and natural flooding including creation of floodplain meadows

Some plans are more developed than others (Lugg and Ithon) whilst others are simply generic checklists, and the Llynfi sub-catchment falls into the latter category. The absence of a Llynfi plan is reviewed below in the context of actions taken in recent years.

NRW supports several surveillance sites within the Wye catchment, all of which have co-located water quality and biological data. **Llangorse Lake**, which historically suffered from severe sewage pollution, is a surveillance site and the only action included in the Llynfi sub-catchment NMP is to:

- reduce nutrient inputs into Llangorse Lake
- restore the inflow and outflow to encourage nutrient deposition on flood zones
- to discuss next steps, whatever that means, at the Technical Advisory Group

In the context of this report, and the WSA's two recent phosphate monitoring reports (*Gwent Angling Society, 2021a, 2021b*), this action appears to be seriously out of step with pollution problems in the catchment and SAC phosphorus compliance failure in the three tributaries.

The Farming Connect Advisory Service provides technical advice in Wales on a range of subjects including infrastructure and nutrient management to farmers, frequently following NRW inspections. Between 2018 and 2020, and focussing on nutrient and soil management, Farming Connect delivered a targeted pollution prevention campaign in priority WFD failing waterbodies, including the Llynfi and Ennig. This included holding on-farm events, hosting local discussion groups, producing e learning, fact sheets and a video on pollution sources and prevention. What impact this had in reducing phosphate pollution *in the three Llynfi sub-catchments*, how it was targeted in relation to specific perceived problems, and how its impact if any was measured, is unknown.

In terms of its quantitative impact in reducing agri-pollution, the findings in this report suggest that the Farming Connect project was not effective. Whether any such work continues in the Llynfi catchment, and how it is targeted, and success measured, is unknown.

Also, the Wye and Usk Foundation's WISE (Wye, Ithon, Severn Ecosystems) project ran until June 2021, with plans to visit at least 275 farms covering 41,000 hectares of agricultural land. Whether any work was done in the Llynfi catchment, and whether it had any material impact, is unknown.

This report and the results from the WSA's continuing nutrient monitoring programme, should provide both evidence and a catalyst to create a sub-catchment plan and timetable for improving water quality in the Llynfi catchment, and foster greater targeted, constructive collaboration between NRW, citizen scientists and stakeholders and conclusions and recommendations are summarised below.

Conclusion

Compliance

- NRW reported (*Hatton-Ellis & Jones, 2021*) that annual average RP levels in the Llynfi catchment during 2017-19 were typically **3-4 times the EQS target** for the respective water bodies.
- WSA measurements taken during the period July 2020 – August 2021 confirm the same picture.
- *Goldsmith et al. (2018)* reported similar phosphorus levels in Llangorse Lake; moreover, the situation at Llangorse is substantially unchanged since the earlier work of *May et al. (2008)*.
- Summary:
 - The entire Llynfi catchment is heavily nutrient-enriched; annual average RP levels are in the region of 0.050 – 0.080 mgP/l, compared to EQS targets of 0.015 – 0.025 mgP/l.
 - Llangorse has severe blue-green algal blooms every year (cyanobacterium)
 - **It has been this way for decades.**

Historical mitigation

- Several projects have been carried out over the years by DC-WW to divert sewage discharges from the upper sub catchment. Sewage treatment in the lower sub catchment has benefitted from the addition of P-stripping at Talgarth STW in 2015.
- *May et al. (2008)* proposed several measures to reduce nutrient emissions on Llangorse Lake: catchment sensitive farming, the establishment of an NVZ and improved management of septic tanks. An NVZ was indeed established in 2012, but it is not known by WSA if any other proposals were acted upon. Self-evidently there has been no subsequent reduction. Indeed, it seems that any reductions secured by DC-WW may have been offset by increases from other sources.
- Historical mitigation is thus heavily focussed around Llangorse Lake. It is not clear what, if any, mitigation measures are in place today within the wider catchment.

Source apportionment model

- We present a simple phosphorus source apportionment model for the Llynfi catchment. We make no attempt to model in-river processes (metabolism, silt adsorption, etc.). Moreover, some of our load calculations are based on first-order estimates of populations etc. Nevertheless, the model aligns convincingly with available water quality measurements.
- Considering the rural sub catchments above the Ennig/Llynfi confluence (Llynfi, Dulas & Ennig):
 - 89% of the P load is from diffuse agricultural sources, 11% is from septic tanks
 - There are no major point sources
- Considering the full catchment:
 - There is an additional P load contribution of 9% from the STWs and PTPs in the lower sub catchment around Talgarth, Bronllys & Three Cocks
 - This reduces diffuse agriculture to 81% and septic tanks to 10% overall
- Hence, on an annual average basis, the P load within Llynfi catchment is dominated by agriculture. In the upper sub catchments, there is very little else there other than the seasonal tourist load centred on Llangorse. We have been unable to include specifically for camping and caravan sites in our model, largely because it is unclear how much of their sewage is routed to Talgarth STW and how much remains on septic tanks. Therefore, it is possible that our septic tank estimate is too low in the sub catchment of the Llynfi above the Dulas confluence. But such an error would not be sufficient to alter the broad conclusion that the dominant source of phosphorus is diffuse agricultural pollution. Nevertheless, it is regrettable that septic tanks appear to be so poorly documented, despite a requirement for registration with NRW.

- Our model shows that three fixed-flow point sources (Talgarth, Aberllynfi & Felindre STWs) become significant at low summer flows. Although our calculations relate to phosphorus load, we note that all other effluent components will have similarly reduced dilution. The AMP8 proposals for additional P-stripping at Talgarth and Aberllynfi STWs should provide major improvement to in-river P concentrations at low flow. However, the Felindre Brook downstream of its STW remains of concern.
- We have examined 2020 CSO emissions within our analysis. Whilst not significantly impacting annual average statistics, they could easily represent significant point pollution events, particularly if occurring during low summer flow conditions. Sadly, published CSO spill data does not include date of occurrence or volume of discharge, so we can only guess the environmental impacts.
 - 4 CSOs are clearly in need of investigation by DC-WW: Trefecca Old Railway, Talgarth STW, Bronllys Garage A479, Three Cocks Pumping Station
- Given that the catchment is diffuse source dominated, it is helpful to look at the components within this category. Figure 9 shows us that 78% of the catchment P-load comes from pasture and associated livestock - of which $\frac{3}{4}$ is sheep and $\frac{1}{4}$ is cattle; only 11% comes from arable farming.

If the aim is to fix a phosphorus overload of 3-4 times target levels, this will not be achieved by tinkering around the edges, for example Farming Connect's (and possibly WUF's) seemingly ineffectual activities in the past. Big changes are needed in order to deliver a big effect. The only meaningful solutions to the Llynfi catchment's phosphorus problem involve paradigm shifts in agriculture, such as:

- Substantial reduction in stock density (say, to pre-war levels)
- Substantial reduction in pasture nutrient levels
- Substantial re-wilding of pasture

It is not for us to tell agriculture how best to go about fixing their problem, but this is clearly way beyond catchment sensitive farming and a few farm visits. This is about radically changing financial incentives. The post-Brexit agricultural grant arrangements offer a golden opportunity to achieve something good for the environment. A model which privatises profit and socialises concomitant pollution cannot be allowed to continue. Manufacturing industry was forced to stop doing this half a century ago. Now agriculture must also step up.

Load apportionment model

- The higher sampling frequencies and improved spatial granularity associated with citizen science data not only allow for improved confidence in compliance assessment but offer an opportunity for more detailed analysis. Load apportionment modelling is one applicable technique which we have applied for phosphorus load at two locations:
 - Pipton Bridge on the lower river, 1.8km from the Wye confluence:
here, the LAM analysis shows a river dominated by diffuse sources for 80% of the year, but with fixed-flow sources (sewage) dominating for the remaining 20% (at low flow rates).
The coefficient derived for the fixed-flow component is an excellent match for the expected phosphorus emissions from Talgarth and Felindre STWs.
 - Llynfi Bridge, 1.1km below the Llangorse Lake outflow:
again, we see a river dominated by diffuse sources for approximately half of the year. However, there is also a fixed-flow signal which dominates at lower flows. The

magnitude of this is equivalent to the output of an STW serving 400 people. Its source is not known, but it appears to originate somewhere within the 900m of river above the bridge.

The source of this pollution needs to be investigated.

In summary, LAM appears to offer a very powerful analysis methodology, which in future could be applied to other determinands such as ammoniacal nitrogen and TDS.

Legacy phosphorus

- We have estimated legacy phosphorus, based on some simple assumptions and the available data on soil Olsen-P. Whilst not perfect, this first-order approximation is better than nothing.
- Analysis of the Wye catchment by the University of Lancaster (*River Wye Nutrient Management Board, January 2021*) suggests that soil phosphorus is currently increasing rapidly, with 34% more phosphorus being applied to soils (predominantly as livestock manure) than is required for crop off-take. This is likely to be applicable to the Llynfi catchment.
- Not only do the phosphorus flows require re-balancing, but to reduce phosphorus export to the river, it is necessary to draw down the current excess. Ultimately, only crop offtake can significantly reduce soil phosphorus levels. Export rates to the river are an order of magnitude less than crop offtake rates. We have estimated the timescale to draw down excessive soil phosphorus to be of the order of ten years; but this largely depends on the starting point (currently unknown) and the desired endpoint (currently undefined).
- The most recent comprehensive Olsen-P survey appears to be for 2007. Given that soil phosphorus concentration is a driving factor for many of the pathways of phosphorus export, it would seem to be an obvious thing to measure and attempt to control. In England, there has been a legal requirement for farmers to measure Olsen-P since 2018. In Wales, such measurements remain best practice guidance only. Nevertheless, if phosphorus pollution of Welsh rivers is to be addressed, then this fundamental lack of data needs to be rectified, and NRW are surely the organisation best placed to collate that data.

Future mitigation and the NMP

- The *generic measures* proposed within the River Wye SAC Nutrient Management Plan are all perfectly sound and reasonable. They represent a reasonable approach for a marginally failing river. But it is difficult to imagine that they could deliver a factor of 3-4 improvement as required for the Llynfi catchment. They are precisely what we mean by *tinkering round the edges* above.
- None of the *generic measures* appears to address the elephant in the room. Current livestock numbers are too high and are only enabled by unsustainable application rates of fertiliser. The resulting phosphorus export rate is more than the river can carry. Furthermore, this aspect does not seem to be adequately dealt with in the '*Long list of practical measures for phosphate reduction*' within the NMP Phosphate Action Plan. If you want a big effect, then start with the big bits. Look at Figure 9.
- It is impossible to dispute the merits of actions to increase baseflow, or deal with specific point sources. But measures to throttle phosphorus pathways and introduce buffering are likely to have limited impact. These measures are precisely what the environmental agencies have been advocating – and presumably doing – for years. Thus far, this approach hasn't worked. If it had, the Wye SAC would not be in such a serious state of decline. So, why would anyone expect a different outcome this time?

- The real danger is that such measures create an illusion of credible action which ultimately fails to deliver – again. Meanwhile this takes the pressure away from the political action which is really needed to solve this problem. The new agricultural grant systems being developed post-Brexit offer a unique opportunity to change financial incentives for the farming industry. As a “simple Welsh catchment” exemplar, this report illustrates how, without fundamental change, the agricultural diffuse phosphorus pollution problem in Wales is intractable, and investment of funding in Farming Connect and WUF’s farm visits, a waste of money with Welsh Government paying lip service to addressing the problem. NRW must now speak truth to power.
- As of September 2021, the Llynfi sub catchment plan has no specific actions proposed. This is disappointing, given that the document has been in preparation throughout 2021, and that the SAC failure was apparent in NRW data from 2017-19. Our load apportionment model estimated that the Llynfi carries an average TP load of 30kgP/day. This needs to be reduced to below 10 kgP/day. What is required is a plan with a credible budget against each action and a means of measuring or estimating progress. And it must add up to at least 20kgP/day.
- We note with interest the ad hoc Herefordshire Farmers’ group, coordinated by Kate Speke-Adams of WUF. This was initiated by farmers concerned at the decline in the River Wye, in full acknowledgement of their contribution to the problems. This has gathered considerable momentum and within just a few weeks has (a) linked into the Avara initiative on poultry manure; (b) banned the application of sewage cake; (c) banned the application of poultry manure, unless indicated by soil nutrient measurements. Something similar in the Llynfi catchment could only be a good thing.
- NRW’s response to the evidence presented in this report requires that analogous, appropriate, concrete steps be taken particularly to address and resolve the key issues of diffuse agricultural pollution and legacy phosphorus, along with a clear timetable for delivery and means of monitoring success.

Recommendations

WSA

- Expand CS measurement locations & optimise sampling strategies.
- Expand EC and ammoniacal nitrogen sampling.
both above now implemented, as discussed in Appendix 1.
- It would be helpful to have lab determination of TP at key locations within the catchment at different times of the year in order to better understand the RP/TP ratio.

DC-WW

- Investigate 4 CSOs as listed earlier.
- Clarify permit status of Llynfi Bridge PS CSO.
- Publish CSO spill times and, where available, volumes on website.
- Publish current STW data (input PE, P-stripping limits, etc.) and future plans on website.
... why does basic information like this need to be an FoI request?

NRW

- Engage with Welsh Government and make plain that fundamental changes to agricultural practices within the Llynfi catchment are needed to achieve SAC EQS compliance.
- Produce a credible sub catchment plan for the Llynfi that builds on this report and the two GAS reports (*Gwent Angling Society 2021a, 2021b*) and reflects the NMP checklist but adds accountabilities, timetables, means of monitoring progress and success in meeting defined

objectives, with potential to modify strategy when and where needed.

This plan needs to be proportionate to the task of reducing annual average RP concentrations by a factor of 3-4. Over and above the measures currently listed in the *Wye SAC Nutrient Management Plan*, it should include:

- An agreed, environmentally sustainable set of targets for soil Olsen-P within the catchment.
- A strategy to monitor soil Olsen-P across the catchment.
- A strategy to draw down soil phosphorus to this target level in a timely manner.
- A strategy to maintain the target level by ensuring that future nutrient application is strictly in balance with crop needs.
- Clarify how CS will engage in that process.
- Consider using this report as a template to apply to other Wye sub catchments. Additionally, consider making the existing evidential basis for improvement plans (SAGIS?) transparent to citizen scientists.
- Encourage Welsh Government to make measurement of Olsen-P a legal requirement for farms in Wales, as it has been in England since 2018; establish systems to collate this data.
- Provide information on current mitigation actions within the Llynfi catchment.
- Investigate Llynfi Bridge fixed flow signal.

To assist with further CS analysis:

- Clarify knowledge of septic tanks and PTPs within the catchment.
- Publish EQS compliance data in real time please, as EA do on their website.
- Update NRFA volumetric flow data in real time please, as EA do.

... once again, why does basic information needs to be an FoI request?

Appendix 1: Future WSA monitoring programme

In the light of this and other analyses of WSA citizen science water quality measurement data, the future monitoring programme has been improved. Guiding principles emerging from this work are:

- Expansion to include all key tributaries, as a minimum to sample immediately above confluences; this allows a rational breakdown into sub catchments.
- Sample frequency at these key points should ideally be once per week, as a minimum; LAM has been shown to be a powerful tool but requires a dataset of at least 25 and preferably 50 samples to provide good statistical significance.
- Sampling should be at a random time of day; many point sources of pollution, in particular sewage related ones, are known to have a diurnal pattern.

Resulting WSA sampling locations are as follow:

River	Location name	Status	OS Grid Ref.
Llynfi	Bridge, Llangorse Car Park	Existing	SO126272
Llynfi	Llynfi Bridge Llangorse	Existing	SO127280
Llynfi	Above Dulas Talgarth	Existing	SO149342
Llynfi	Sewage Works Talgarth	Existing	SO150347
Felindre	Road Bridge Felindre	Existing	SO186365
Llynfi	Pipton Bridge	Existing	SO167379
Dulas	Dulas, above Tryffrwd confluence	New	SO119339
Tryffrwd	Tryffrwd above Dulas confluence	New	SO126343
Dulas	Dulas, Above Llynfi Confluence	New	SO146344
Ennig	Ennig, Arosfa	New	SO173326
Ennig	Ennig, Middle Genfford	New	SO170314
Ennig	Allotment Bridge Talgarth	Existing	SO151344

Table 34: WSA sampling locations within the Llynfi catchment

Additional volunteers have been recruited to support this enlarged programme. Locations are shown on the map overleaf. In addition to temperature and phosphate, electrical conductivity (EC) is also being recorded at eleven of the twelve locations, plus ammoniacal nitrogen and pH at two of these.

It should be noted that several of these locations are also monitored by NRW, affording direct comparisons to be made with a data set significantly larger than NRW's.

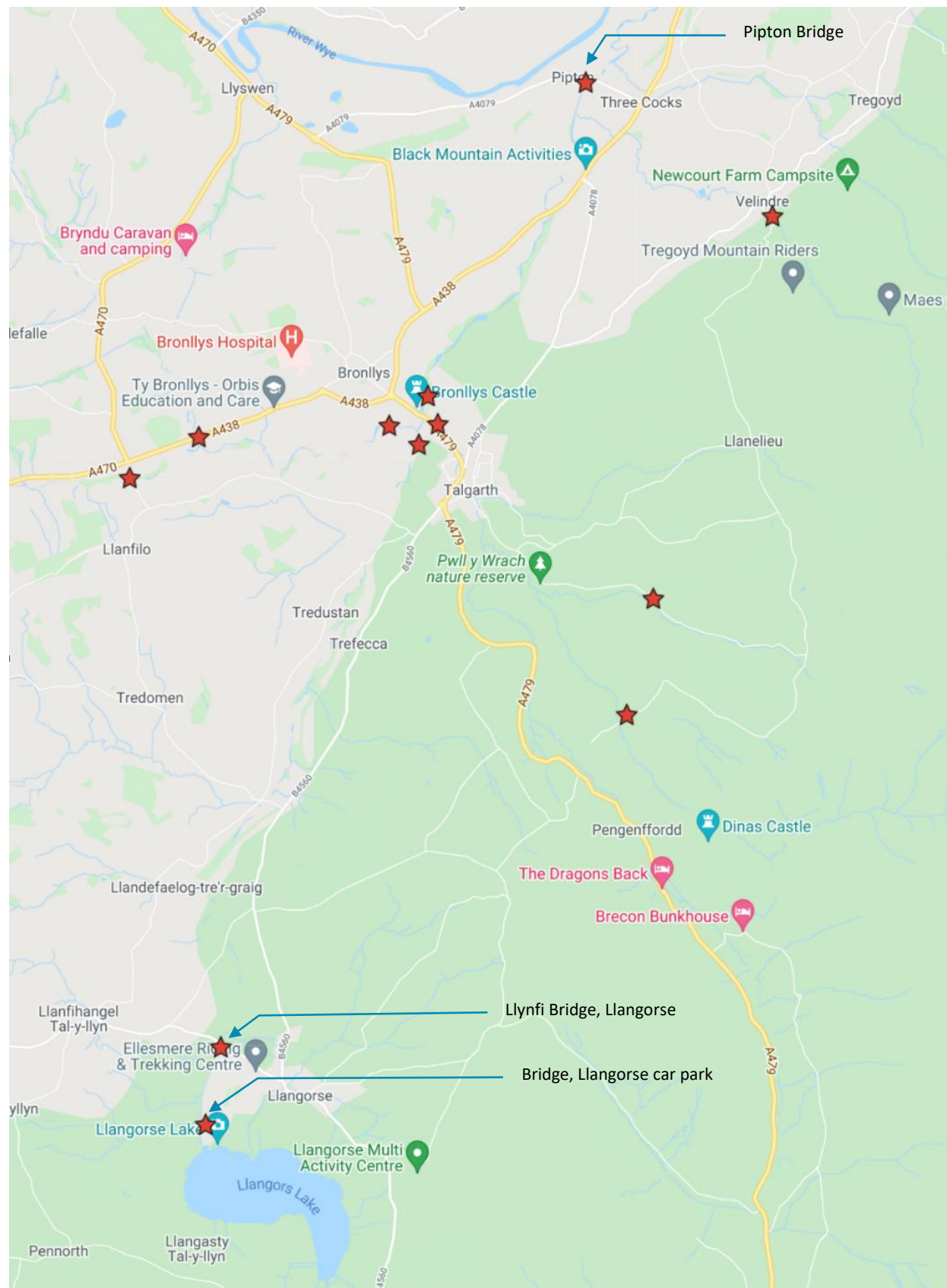


Figure 12: Map showing WSA water quality sampling locations (red stars); the three locations referenced in this report are indicated

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List of abbreviations

BOD	Biochemical Oxygen Demand
ca	Capita (head of livestock)
CEH	Centre for Ecology & Hydrology
CPRW	Campaign for the Protection of Rural Wales
CSO	Combined Storm Overflow
DC-WW	Dwr Cymru – Welsh Water
EC	Electrical Conductivity
EQS	Environmental Quality Standard
FoI	Freedom of Information
GAS	Gwent Angling Society
LAM	Load Apportionment Modelling
NMP	Nutrient Management Plan
NRFA	National River Flow Archive
NRW	Natural Resources Wales
PE	Population Equivalent
PTP	Package Treatment Plant
RP	Reactive Phosphorus
SAC	Special Area of Conservation
SAM	Source Apportionment Modelling
SRP	Soluble Reactive Phosphorus
ST	Septic Tank
STW	Sewage Treatment Works
TDS	Total Dissolved Solids
TP	Total Phosphorus
UKSO	United Kingdom Soil Observatory
WFD	Water Framework Directive
WUF	Wye and Usk Foundation